BASIC CIVIL & MECHANICAL ENGINEERING

Subject Code: 23A01201T | UNIT – 1 : BASICS OF CIVIL ENGINEERING

1. Role of Civil Engineers in Society

> **Civil Engineering** is the oldest branch of engineering – the art and science of **planning, designing, constructing, operating and maintaining** the built and natural environment: buildings, roads, bridges, dams, canals, water supply, sewers, airports, ports, railways, tunnels.

- Provides the **basic infrastructure** needed for human life – shelter, safe water, sanitation, transport, power projects, irrigation.
- Contributes to **economic development** – good roads, ports and railways reduce cost of goods and create jobs.
- Ensures **safety** of structures against load, earthquake, wind, flood & fire.
- Protects the **environment** – water/ waste-water treatment, solid waste management, pollution control.
- Plans **smart cities**, housing for all, disaster-resistant & sustainable construction.
- Famous Indian civil engineer: **Sir M. Visvesvaraya** (Krishnaraja Sagara dam) – **Engineers' Day = 15 September.**

❖Responsibilities of a civil engineer

- Survey & investigation → planning → design → estimating & costing → construction supervision → quality control → maintenance.
- Works with architects, contractors, mechanical, electrical engineers & government.

> Civil engineer = builder of the nation's infrastructure

2. Various Disciplines of Civil Engineering

- Structural Engineering
- Geo-technical Engineering
- Transportation Engineering
- Hydraulics & Water Resources Engineering

- Environmental Engineering
- (Also: Construction Management, Surveying/ Geomatics, Coastal & Earthquake Engineering)

❖(i) Structural Engineering

- Deals with **analysis and design of structures** so that they safely resist loads (dead, live, wind, earthquake, snow).
- **Scope:** RCC and steel buildings, bridges, towers, chimneys, industrial sheds, water tanks, dams, high-rise & tall structures, offshore platforms, earthquake-resistant & prestressed structures.
- Uses: structural analysis software (STAAD, ETABS), IS codes (IS 456 for concrete, IS 800 for steel, IS 1893 for earthquake).
- Concerned with **strength, stability, serviceability, economy.**

❖(ii) Geo-technical Engineering

- Study of **soil and rock** behaviour and their use as foundation or construction material.
- **Scope:** soil investigation (boring, SPT), **bearing capacity**, design of **foundations** (shallow – footing, raft; deep – piles, wells), **retaining walls**, slope stability, embankments, earth dams, ground improvement, tunnelling, landslide control.
- Soil properties: grain size, plasticity (Atterberg limits), permeability, shear strength, compaction, consolidation.

❖(iii) Transportation Engineering

- Planning, design, construction & operation of **highways, railways, airports, harbours, tunnels, pipelines.**
- **Scope:** traffic engineering, pavement design, highway geometric design, road safety, urban transport (metro, BRTS), railway track design, airport layout, port planning.
- Reduces travel time, cost, accidents and pollution.

❖(iv) Hydraulics & Water Resources Engineering

- Study of **water flow, storage, distribution and control.**
- **Scope:** hydrology, **dams & reservoirs, canals, irrigation**, hydropower, flood control, river training, groundwater (wells), water supply schemes, drainage, pumps & turbines, ports/ coastal protection.

- Basic principles: continuity, Bernoulli's equation, flow in pipes & open channels.

❖(v) Environmental Engineering

- Protection of environment & public health.
- Scope: water supply & treatment, **waste-water (sewage) treatment**, solid-waste management, air & noise pollution control, **environmental impact assessment (EIA)**, hazardous & e-waste, climate change, green buildings, sustainable development.

3. Building Construction & Planning

❖Components of a building

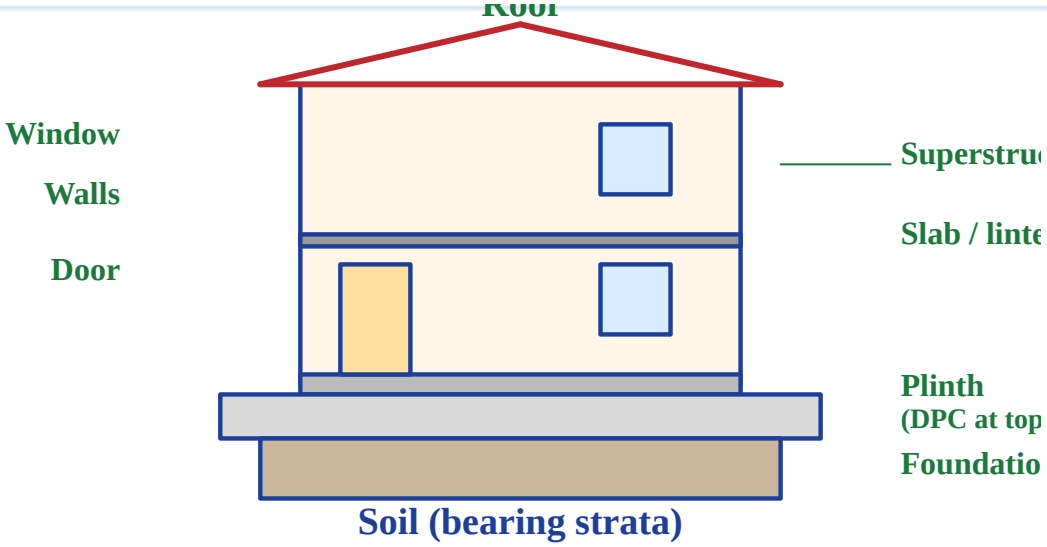


Fig 1.1 Components of a building: sub-structure (below plinth) and super-structure (above)

- **Sub-structure (below ground/ plinth): foundation** – transfers load safely to soil; **plinth** – raised portion above ground level (≥ 450 mm) with **DPC (damp-proof course)** to stop capillary moisture.
- **Super-structure:** walls (load bearing/ partition), columns, beams, slabs, lintels, sills, doors, windows, staircases, roof, floors, finishing (plaster, paint).
- **Foundation types:** isolated footing, combined footing, strip, **raft**, pile, well.
- **Walls** are built with brick/ stone/ concrete blocks in **bonds (English, Flemish)** to avoid vertical continuous joints.
- **Lintel:** beam over openings (doors/ windows). **Sill:** bottom of a window. **Chajja (sunshade).**

❖Types of buildings (by use)

- Residential, commercial, educational, institutional (hospital), industrial, assembly (cinema, auditorium), storage.

❖Types of construction

- **Load-bearing structure** – walls carry loads (up to 3 floors). **Framed (RCC) structure** – columns & beams carry loads; walls are only partitions (high rise).
- **Composite** – steel + concrete.

4. Building Planning

> **Planning** = arranging various rooms/ spaces of a building in a suitable manner to satisfy the needs of the occupant, economically, as per rules (building bye-laws).

❖Principles of planning

- **Aspect:** orientation of rooms to get sun, air (bedroom – east/ south-east; kitchen – south-east; living – north/ east; toilet – north-west).
- **Prospect:** pleasing outside view. **Privacy** – from neighbours & within house.
- **Grouping:** related rooms close – kitchen near dining; bedroom with attached toilet.
- **Roominess:** better space utilisation; correct size of rooms.
- **Furniture requirement, sanitation, circulation** (corridors, staircases), **lighting & ventilation** (window area ≥ 1/10 floor area), **flexibility, economy, elegance, practicability.**
- Follow **local bye-laws:** FAR/ FSI, **setbacks**, plot coverage, height limit.

FSI (FAR) = Total built-up floor area of all floors / Plot area

- **Minimum room sizes (NBC):** habitable room ≥ 9.5 m^2 (min width 2.4 m); kitchen ≥ 5 m^2; bathroom ≥ 1.8 m^2; ceiling height ≥ 2.75 m.
- **Drawings:** plan, elevation, section, site plan, foundation plan; scale 1:100.
- **Vaastu / Green building** and **rainwater harvesting**, solar water heater, STP are now included.

5. Construction Materials

- Materials used: stone, brick, cement, sand, aggregate, timber, concrete, steel, glass, bitumen, paints, tiles, plastics, composites.
- Selection: strength, durability, cost, availability, workability, appearance, thermal & fire resistance.

❖5.1 Cement

> Cement is a fine, grey, binding material that sets and hardens when mixed with water and binds aggregates together. Invented by Joseph Aspdin (1824) – "Portland cement".

- Raw materials: limestone ($CaCO_3$, calcareous) + clay/ shale (argillaceous – silica, alumina) + gypsum (3–5%) added at the end.
- Manufacture (wet or dry process): crush → grind → mix (slurry) → burn in rotary kiln at 1400–1500 °C → clinker → cool → grind with gypsum → bag (50 kg).

❖Main compounds (Bogue)

Compound	Formula	%	Role
Tricalcium silicate (C_3S)	$3CaO·SiO_2$	40–50	Early strength, heat
Dicalcium silicate (C_2S)	$2CaO·SiO_2$	25–30	Later strength
Tricalcium aluminate (C_3A)	$3CaO·Al_2O_3$	5–10	Flash set; gypsum controls it
Tetra-calcium alumino-ferrite (C_4AF)	$4CaO·Al_2O_3Fe_2O_3$	10	Colour

❖Types of cement (BIS)

- OPC (Ordinary Portland Cement) 33, 43, 53 grade – (28-day compressive strength 33/43/53 N/mm^2); IS 269 / 8112 / 12269.
- PPC (Portland Pozzolana Cement) – 15–35% fly ash; low heat, good in marine, cheaper – IS 1489.
- PSC (Slag cement), Rapid hardening, Low heat, Sulphate-resisting, White cement, Hydrophobic, High alumina.

❖Properties & tests

- Fineness (≥ 225 m^2/kg, sieve test), Consistency (Vicat – 26–33% water), Setting time: initial ≥ 30 min, final ≤ 600 min (10 h), Soundness (Le-Chatelier – expansion ≤ 10 mm), Compressive strength of 70.6 mm cement mortar cubes (3, 7, 28 days), specific gravity 3.15.
- Uses: mortar, plaster, concrete, grouting, masonry, flooring.
- Store in dry place, off the ground, away from walls, in stacks ≤ 10 bags; use within 3 months.

❖5.2 Aggregates

> Inert granular materials (sand, gravel, crushed stone) which form 60–75% of concrete volume.

- Fine aggregate: passes 4.75 mm IS sieve, retained on 75 µm (sand; river sand, M-sand, crushed sand). Zones I–IV.
- Coarse aggregate: retained on 4.75 mm sieve – gravel, crushed granite/ basalt; sizes 10, 12.5, 20, 40 mm (20 mm normally for RCC).
- All-in aggregate = fine + coarse together.

❖Required properties

- Hard, strong, durable, clean (free of clay, silt, organic matter, salts), angular & rough-textured (good bond), well graded (smallest voids), specific gravity 2.6–2.8, low water absorption (< 2%).
- Tests: sieve analysis (fineness modulus), crushing value, impact value, abrasion (Los Angeles), specific gravity, bulking (sand swells 20–40% at 5–8% moisture), flakiness/ elongation index.
- Well-graded aggregates → dense, strong, economical concrete (less cement).

❖5.3 Bricks

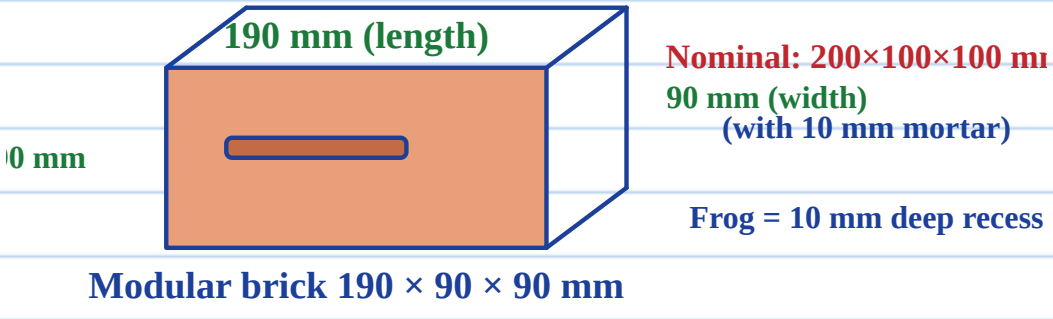


Fig 1.2 Standard (modular) brick

> **Brick** = a rectangular building unit made by moulding, drying and burning clay. Most popular wall material.

- **Standard (modular) size:** 190 × 90 × 90 mm (nominal 200 × 100 × 100 mm with mortar). Traditional brick 230 × 110 × 70 mm. **Weight** ≈ 3 kg.
- **Composition of good brick earth:** silica 50–60% (prevents shrinking), alumina 20–30% (plasticity), lime < 5%, iron oxide 5–6% (red colour, fusion), magnesia < 1%.
- **Manufacture:** (1) **preparation of clay** – digging, weathering, tempering; (2) **moulding** (hand/ machine); (3) **drying** (3–10 days); (4) **burning** in clamp/ Hoffmann kiln at 900–1100 °C.

❖Classification (IS 1077)

- **First class** – table-moulded, uniform shape, deep red, compressive strength > 10.5 N/mm^2 (class 10+), absorption ≤ 20% – for superstructure.
- **Second class** – ground-moulded, slightly distorted, absorption up to 22%; for plastered walls.
- **Third class** – under-burnt, soft, absorption high (25%) – temporary works.
- **Fourth class / jhama** – over-burnt, vitrified; crushed to coarse aggregate.
- **Others:** fly-ash bricks, hollow/ perforated, fire bricks, AAC blocks, concrete blocks, sand-lime bricks.

❖Qualities of good brick

- Uniform size & shape, sharp edges, **metallic ringing sound** when struck, no cracks, scratch-resistant, uniform colour, **absorbs ≤ 20% water in 24 h**, no efflorescence (white salt), compressive strength ≥ 3.5 – 35 N/mm^2.

❖5.4 Cement Concrete

> **Concrete** = Cement + Fine aggregate + Coarse aggregate + Water (+ admixtures). **Plain cement concrete (PCC)** is strong in compression but weak in tension; **RCC** = concrete + steel bars.

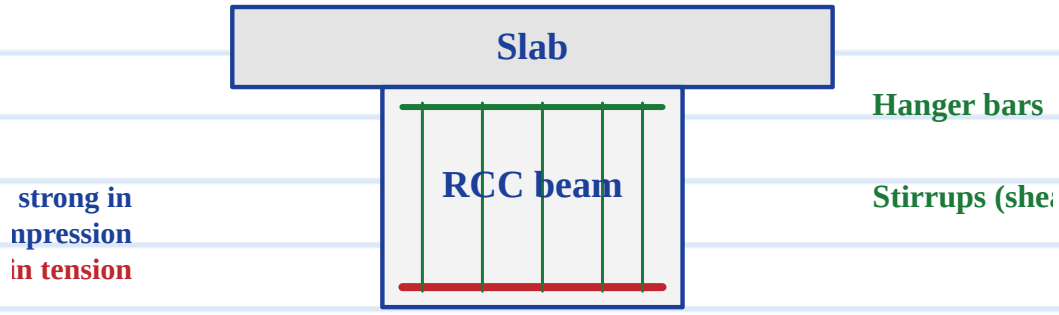


Fig 1.3 Reinforced cement concrete (RCC) beam section

❖Grades of concrete (IS 456)

- M20 (1 : 1.5 : 3) – "M" = mix; 20 = characteristic compressive strength (N/mm^2) at 28 days on 150 mm cubes. Other grades: M10, M15, M20, M25, M30 … M80.

Grade	Ratio (C:S:A)	Use
M5	1 : 5 : 10	Leveling/ PCC under foundation
M10	1 : 3 : 6	PCC base
M15	1 : 2 : 4	Plain concrete
M20	1 : 1.5 : 3	RCC – general
M25	1 : 1 : 2	RCC – heavy duty

❖Properties

- **Workability** (slump test: 25–100 mm; compacting factor), **w/c ratio** (0.4–0.6; lower ratio = higher strength, **Abram's law**), **strength** (compressive – cube test), **durability**, **impermeability**, **shrinkage**, **creep**, density 2400 kg/m^3, RCC 2500 kg/m^3.

❖Steps in concreting

- **Batching** (weigh/ volume) → **mixing** (machine ≥ 2 min) → **transporting** → **placing** → **compaction** (needle vibrator – removes air voids) → **finishing** → **curing** (keep moist 7–28 days; strength gain, avoids cracks).
- **Cement : water** – 1 bag (50 kg) cement needs ≈ 25–30 ℓ water.
- **Admixtures:** plasticizers, retarders, accelerators, air-entraining agents; **mineral:** fly ash, silica fume.

❖Advantages & drawbacks

- Moldable, durable, fire-resistant, economical, high compressive strength; but heavy, weak in tension, needs curing, cracks with shrinkage.

❖S.S Steel

> Steel = alloy of iron + carbon (0.05–1.5%). Mild steel (0.15–0.25% C) is the most used in construction; strong in both tension and compression.

- Reinforcement bars (rebars): plain mild steel (Fe 250), TMT (Thermo-Mechanically Treated) bars – Fe 415, Fe 500, Fe 550; ribbed surface for bond; diameter 6–40 mm.
- Structural steel sections: I-section (ISMB), channel (ISMC), angle (ISA), T-section, flats, plates, hollow tubes; used in trusses, steel buildings, bridges, towers, industrial sheds.
- Properties: high strength, ductility, toughness, weldable, 100% recyclable; density 7850 kg/m^3, Young's modulus 2×10^5 N/mm^2, yield stress 250 N/mm^2 (Fe 250).
- Stress–strain: elastic limit → yield point → plastic → ultimate → rupture.
- Disadvantages: corrosion (needs paint/ galvanising), loses strength in fire (above 550 °C), costly.
- Corrosion protection: painting, galvanising, epoxy-coated bars, cathodic protection.
- Other materials (brief): timber (renewable, doors/ windows), glass, bitumen, tiles, aluminium, FRP / composites, geosynthetics.

6. Prefabricated Construction Techniques

> Prefabrication = manufacturing building components (walls, slabs, beams, columns, staircases, even whole rooms) in a factory or casting yard, transporting them to site and assembling/ erecting them.

❖Features of the technique

- Components are standardised, cast in moulds under controlled conditions; site work is only joining (dry/ wet joints, bolting, grouting).
- Types: precast concrete (RCC/ prestressed) panels, hollow-core slabs, precast walls, steel (pre-engineered) buildings, modular/ volumetric units (container-type), 3D printed walls.

❖Advantages

- Faster construction (time ↓ 30–50%), better quality control, less site labour, less wastage, less formwork & scaffolding, all-weather production, cost saving in repetition, cleaner site, eco-friendly.

❖Disadvantages

- High initial investment (plant, moulds), transport & lifting costs (cranes), joints need care (water leakage, structural continuity), limited design flexibility, skilled labour for erection, economical only for large volume / repetition.

❖Applications

- Mass housing, schools, hospitals, flyovers/ metro viaduct segments, precast piles, railway sleepers, compound walls, culverts, modular toilets, industrial sheds, PMAY (Light House Projects).

> Prefabrication → Factory made + Site assembled = Speed + Quality + Economy

7. Quick Revision & Important Questions

Material	Key facts
Cement	Limestone+clay, kiln 1450 °C, IST ≥ 30 min, FST ≤ 10 h
Brick	190×90×90 mm; absorption ≤ 20 %; > 3.5 N/mm^2
Concrete	C+S+A+W; M20 = 1:1.5:3; cure 28 d
Steel	Fe 500 TMT; density 7850 kg/m^3

- Define civil engineering; explain its role in society.
- Write about the various disciplines of civil engineering and scope of each.
- List the components of a building; explain building planning principles.
- Write notes on cement, aggregates, bricks, concrete and steel.
- What is prefabrication? State its advantages and disadvantages.

8. Additional Topics for Better Marks

❖Mortar

- Mortar = binder (cement/ lime) + sand + water – used to join bricks/ stones and for plaster. Cement mortar 1:4 for brickwork; 1:6 for plaster; 1:3 for waterproof work.

❖Bonds in brick masonry

- **Stretcher** (long side seen) – 1/2 brick wall; **Header** (short side seen) – full-brick wall; **English bond** (alternate stretcher & header courses – strongest); **Flemish bond** (alternate header-stretcher in each course – good appearance).
- Rules: no continuous vertical joints, use **closers/ queen closers**, joints 10 mm, wet bricks before laying.

❖Timber, glass and others (brief)

- **Timber (teak, sal, deodar):** doors, windows, roofs; seasoning removes moisture; weak against fire/ termites. **Plywood, particle board** used in interiors.
- **Glass:** transparent; used for windows, facades; toughened/ laminated/ tinted types.
- **Bitumen:** black binder for roads & waterproofing. **Paints/ distemper** protect surfaces. **Tiles** for floors. **Aluminium** and **PVC** for doors/ windows.

❖Foundations – quick facts

- **Safe bearing capacity (SBC):** soft clay 50–100 kN/m^2; sand 200–400; hard rock > 3000 kN/m^2.
- Depth of foundation ≥ 0.9–1.2 m below ground; width of footing = Load / SBC.

9. Solved Problems

❖Problem 1 – Number of bricks

> Wall 10 m × 3 m × 0.2 m (modular brick with mortar 0.2×0.1×0.1 m = 0.002 m^3). Volume = 6 m^3 → bricks = 6/0.002 = **3000 bricks** (≈ 5% wastage → 3150).

❖Problem 2 – Materials for 1 m^3 of M20 concrete (1 : 1.5 : 3)

> Dry volume = 1.54 m^3. Sum of ratio = 5.5. Cement = 1.54 × 1/5.5 = 0.28 m^3 = 0.28 × 1440 = **403 kg ≈ 8 bags**; Sand = 1.54 × 1.5/5.5 = **0.42 m^3**; Coarse aggregate = 1.54 × 3/5.5 = **0.84 m^3**.

❖Problem 3 – Water for concrete

> Cement 400 kg, w/c = 0.5 → water = **200 litres**.

❖Problem 4 – Plinth area

> Plot 15 m × 20 m = 300 m^2; built-up per floor = 150 m^2; 3 floors = 450 m^2 → FSI = 450/300 = 1.5.

❖Problem 5 – Footing width

> Column load 600 kN; SBC = 150 kN/m^2 → area = 600/150 = 4 m^2 → square footing **2 m × 2 m**.

10. Sketch & Answer-Writing Tips

- Start every answer with a **definition**, give 4–5 clear points, end with **uses/advantages**.
- Draw neat diagrams: building section (label foundation, plinth, DPC, wall, lintel, slab), brick, RCC beam (steel at bottom).
- Mention **IS code numbers** and grades where possible (IS 456, IS 269, IS 1077, IS 1786).

BASIC CIVIL & MECHANICAL ENGINEERING

Subject Code: 23A01201T | UNIT – II : SURVEYING

I. Introduction & Objectives of Surveying

> **Surveying** is the art and science of determining the **relative positions of points on, above or beneath the earth's surface** by measuring horizontal distances, vertical distances (heights) and angles, and plotting them to scale as a **plan or map**.

❖Objectives

- To prepare **plans/ maps** of an area to a suitable scale showing natural & man-made features.
- To determine **areas and volumes** (earthwork, reservoir capacity).
- To **set out** (locate) structures on the ground – buildings, roads, canals, bridges, tunnels.
- To get **levels/ elevations** & contours for design of roads, railways, drainage, dams.
- Fixing **boundaries**, land records, town planning, mining, defence & geological studies.

❖Primary divisions

- **Plane surveying:** earth surface treated as **plane**; area < 260 km^2 – ordinary civil work.
- **Geodetic surveying:** curvature of earth is considered; large areas; high accuracy (national control).

❖Classification (by instruments/ methods)

- **Chain (tape) surveying, Compass surveying, Plane table surveying, Levelling, Theodolite surveying, Tacheometry, Total station / GPS / drone / photogrammetry / remote sensing.**
- By purpose: topographical, cadastral (property), city, engineering, hydrographic, mine, route, astronomical.

❖Fundamental principles

- Working from whole to part (fix big control framework first, then fill in details – prevents error accumulation).
- Locating a new station by at least two measurements from fixed points (linear/ angular).
- Provide checks; take independent measurements; keep field notes neat.

2. Horizontal Measurements – Linear Measurements

- Horizontal distance between two points is measured by direct methods (chain/ tape) or indirect (tacheometry, EDM, total station, GPS).

❖Instruments

- Metric chain: 20 m (100 links, each link 0.2 m) or 30 m; tags every 2 m / 5 m; brass handles. Gunter's chain 66 ft (100 links). Steel band; Invar tape (precise); cloth/ metallic tape (20, 30 m).
- Arrows (marking pins, 10 per set), ranging rods (2 m, painted red-white), offset rod, plumb bob, cross-staff / optical square (to set right angles), pegs, EDM/ laser distance meter.

❖Ranging

> Ranging = fixing intermediate points on a chain line so that the chain is laid in a straight line.

- Direct ranging: by eye with ranging rods when end stations visible. Indirect ranging: by reciprocal ranging when intermediate ground is high.

❖Chaining on level & sloping ground

- Hold the chain horizontal and taut; leader & follower; arrows exchange every chain length; offsets measured perpendicular to the chain line (perpendicular or oblique).
- Slope correction: horizontal length = $L \cos\theta$ (or $C_s = h^2/2L$).

❖Chain survey (simple)

- Area divided into well-conditioned triangles (angles 30°–120°, ideally equilateral), base line, tie lines, check lines; field book records (chainage, offsets).
- Suitable for small, open, flat areas.

❖Errors in chaining & corrections

- Cumulative vs compensating; Instrumental (chain too long/ short), natural (temperature, sag), personal, mistakes (miscounting).
- Correction for incorrect length: True length = $L \times (L'/L)$ where L' = actual length of chain, L = nominal.
- Tape correction: temperature $C_t = \alpha (T_m - T_0) L$; sag $C_s = w^2L^3/(24P^2)$; pull $C_p = (P - P_0)L/(AE)$.

❖Example

> A 20 m chain was found to be 20.05 m long. A line measured 400 m with this chain. True length = 400 × 20.05/20 = **401.0 m**. If the chain is 0.05 m too short: true = 400×19.95/20 = **399.0 m**.

> A line of 150 m on a slope of 5°: horizontal distance = 150 cos 5° = **149.43 m**.

3. Angular Measurements

- Horizontal angles are measured with **compass** (magnetic bearings), **theodolite** (accurate angles), **sextant**, **total station**, and **plane table** (graphically).

❖Theodolite

- Most accurate instrument for **horizontal & vertical angles**; also used for prolonging lines, levelling, tacheometry.
- Parts: telescope, **vertical circle** (graduated), **horizontal circle**, plates, **levelling head** (foot screws, tribrach), plate levels, **vernier/ micrometer**, tripod & plumb bob.
- Transit (vertical-axis, telescope can be rotated 360° in vertical plane); Vernier vs electronic/ digital (least count 20", 10", 1").
- Temporary adjustments: centring, levelling, elimination of parallax.
- Measuring horizontal angle: set up at station O, level; sight left object A (zero), note reading; clockwise to B, read; angle AOB = difference; repeat with face right & take mean to reduce errors (face left & face right observations).

❖Total station

- Electronic theodolite + EDM + microprocessor → gives angles, distance, coordinates (X, Y, Z) simultaneously, stores data; used for large projects, mapping, setting out.
- GPS/ GNSS: satellite based positioning; Drone/ LiDAR surveys for contours/ DEM.

4. Introduction to Bearings

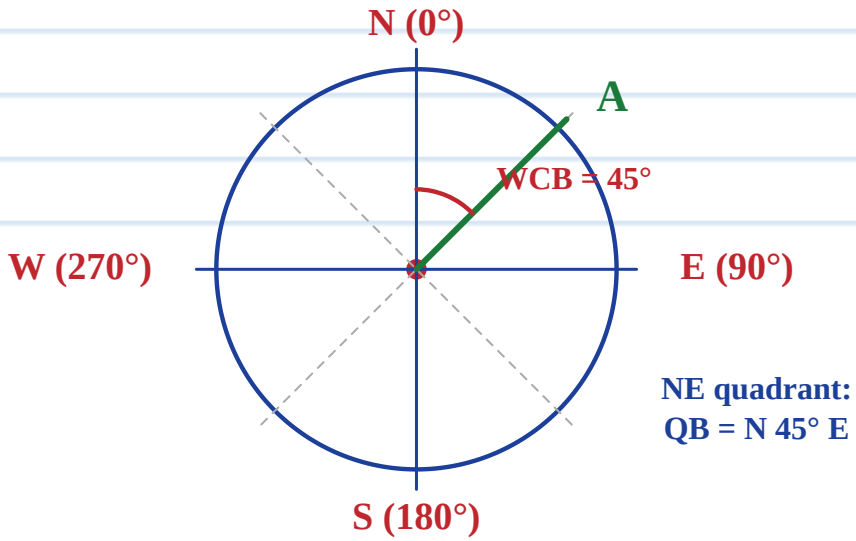


Fig 2.1 Whole circle bearing (WCB, clockwise from North) and quadrantal bearing

> Bearing of a line = horizontal angle between a reference meridian (North) and the line, measured clockwise.

- Meridians: True (geographic north pole), Magnetic (direction of freely suspended needle), Arbitrary.
- Whole Circle Bearing (WCB): measured clockwise 0°–360° from North.
- Reduced / Quadrantal Bearing (RB/ QB): angle from N or S towards E or W, 0°–90° – e.g. N 45° E, S 30° W.

❖Conversion WCB ↔ QB

WCB range	Quadrant	QB
0° – 90°	NE	QB = WCB (N θ E)
90° – 180°	SE	QB = 180° – WCB (S θ E)
180° – 270°	SW	QB = WCB – 180° (S θ W)
270° – 360°	NW	QB = 360° – WCB (N θ W)

❖Fore bearing & Back bearing

> Back bearing (BB) = Fore bearing (FB) ± 180° (+180° if FB < 180°, −180° if FB > 180°)

- FB: bearing in the direction of survey progress; BB: reverse direction.

❖Included angle from bearings

> Included angle at a station = (Back bearing of previous line) ~ (Fore bearing of next line) – i.e. difference between the FB of next and BB of previous line

- Interior angle sum of a closed traverse = (2n − 4) × 90°.

❖Prismatic compass

- Used to measure bearing: **magnetic needle** (balanced on pivot), graduated **aluminium ring/ card** (0° at south end, reading through a prism – WCB), sighting vane (object slit & hair), **prism, lid with mirror**, glass cover, brake pin. Least count 30′.
- **Surveyor's compass:** card fixed to needle; quadrantal bearing; E and W interchanged.

❖Magnetic declination & dip

- **Declination** = angle between true & magnetic meridians (varies place to place & time); **True bearing = Magnetic bearing ± declination (East +).**
- **Dip** = inclination of needle; counter-balanced by a rider (brass coil).

❖Local attraction

- Disturbance of the needle by magnetic substances (iron, steel, power lines). **Detected when FB − BB ≠ 180°. Corrected by the first-line method** or by correction in included angles.

❖Solved problems on bearings

Q1. FB of AB = 62° 30′ → BB = 242° 30′. FB of CD = 215° → BB = 215° − 180° = 35°.

Q2. Convert WCB 285° to QB: in NW quadrant → QB = 360° − 285° = N 75° W. WCB 135° → S 45° E. WCB 215° → S 35° W.

Q3. QB = S 40° E → WCB = 180° − 40° = 140°. QB N 25° W → WCB = 335°.

Q4. FB of AB = 80°, BB of BC (i.e. bearing of CB) = 300°, find ∠ABC. FB of BC = 300° − 180° = 120°; BB of AB = 80° + 180° = 260°. Angle at B = (FB of BC) − (BB of AB) = 120° − 260° = −140° → +360° = 220° (exterior), interior = 140°.

Q5. Magnetic bearing of a line = 125° 30′, declination = 4° 20′ W. True bearing = 125°30′ − 4°20′ = 121° 10′.

Q6. Local attraction: FB of AB = 45° 10′, BB = 226° 30′ → difference 181° 20′ ≠ 180°. Mean = (45°10′ + (226°30′ − 180°))/2 → correct FB = 45° 50′ if station A is free of attraction.

5. Levelling

Levelling is the branch of surveying that determines **elevations (heights)** or **differences in elevation** of points with respect to a **datum**, using a level and staff.

❖Basic terms

- **Datum:** reference surface (**Mean Sea Level – MSL**, Karachi/ Chennai MSL in India).
- **Reduced Level (RL):** elevation of a point above datum. **Benchmark (BM):** fixed point of known RL (GTS BM, permanent, temporary, arbitrary).
- **Line of collimation:** line of sight through the crosshair & optical centre. **Height of instrument (H.I.)** = RL of line of collimation.
- **Back sight (BS):** first reading after setting up level, on a point of known RL. **Fore sight (FS):** last reading before shifting. **Intermediate sight (IS):** any other reading between BS & FS. **Change point (CP / turning point):** point where both FS & BS are taken as the instrument is shifted.
- **Station:** point where staff is held. **Rise & Fall.**

6. Instruments for Levelling

- **Dumpy level:** telescope rigidly fixed to the stand; **bubble tube**, foot screws, **levelling head**, tripod; simplest, most common; adjustments: temporary (setting up, levelling, focusing) and permanent.
- **Tilting level:** telescope can be tilted slightly about a horizontal axis; for quick, precise work.
- **Automatic (self-aligning) level:** compensator keeps line of sight horizontal – fast, popular today.
- **Digital level / laser level / Total station** – bar-coded staff; automatic reading & booking.
- **Y-level, cooke's reversible level** (older types).
- **Levelling staff:** wooden/ aluminium rod, **telescopic or folding**, 3–5 m long, graduated in metres, decimetres, centimetres (E-shaped, 5 mm least count; black & red numerals). Held **vertical** (check by plumb bob/ staff bubble). Foot plate for change points.

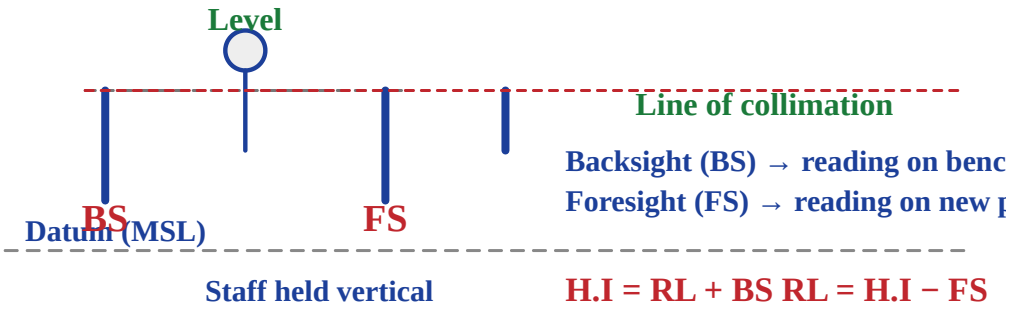


Fig 2.2 Levelling – height of instrument method

❖Temporary adjustments of a dumpy level

- **Setting up** the level (tripod head approx level, firmly fixed). **Levelling up** with 3 foot screws using the plate bubble (two screws parallel to a line, third perpendicular; turn 90°). **Elimination of parallax** (eye-piece focus on cross-hairs, objective focus on staff).

❖Curvature & refraction

- Combined correction = $-0.0673\ d^2$ (d in km, result in m); applied for long sights (> 100–150 m; ignored when BS = FS distances equal – so keep BS and FS distances equal).

7. Booking & Reduction of Levels

❖(a) Height of Instrument (Collimation) method

H.I. = RL of BM + BS ; RL of point = H.I. − FS (or IS) ; New H.I. at change point = RL of CP + new BS

❖(b) Rise & Fall method

- Rise/ Fall = (previous reading) − (current reading); positive = Rise (+), negative = Fall (−). RL = previous RL + Rise or − Fall.
- Arithmetic check: $\Sigma BS - \Sigma FS = \Sigma$ Rise $- \Sigma$ Fall = Last RL − First RL.

❖Which is better?

- HI method – quicker, convenient when many intermediate sights (profile levelling). Rise/Fall – checks every point; used when IS are few.

❖Types of levelling

- Simple (one setup), Differential (many setups between far points), Profile/ longitudinal sectioning (along a road/ canal line at 20–30 m intervals), Cross-sectioning (perpendicular to profile), Reciprocal levelling (across river/ valley – eliminates curvature & refraction), Fly levelling (checking), Check levelling, Precise levelling, Barometric/ trigonometric (indirect).

❖Example 1 – H.I. method

Station	BS	IS	FS	H.I.	RL
BM	1.525			101.525	100.000
A		1.940			99.585
B			2.105		99.420
B (CP)	1.355			100.775	99.420
C			0.885		99.890

RL of A = 101.525 − 1.940 = 99.585; RL B = 101.525 − 2.105 = 99.420; new H.I. = 99.420 + 1.355 = 100.775; RL C = 100.775 − 0.885 = 99.890. Check: $\Sigma BS - \Sigma FS$ = (1.525+1.355) − (2.105+0.885) = −0.110 = 99.890 − 100.000 ✔.

❖Example 2 – Rise & Fall

Readings: 2.285 (BS on BM, RL 200.000), 1.650, 1.895, 2.350 (CP), 1.545 (BS), 0.975 (FS).

Rise 1→2: 2.285−1.650 = +0.635 → RL 200.635; 2→3: 1.650−1.895 = −0.245 → 200.390; 3→CP: 1.895−2.350 = −0.455 → 199.935; after shift: 1.545 − 0.975 = +0.570 → 200.505.

❖Example 3 – Difference in elevation

> BS on A = 1.275, FS on B = 2.610. Difference = 1.275 − 2.610 = −1.335 → B is 1.335 m lower than A.

8. Contouring

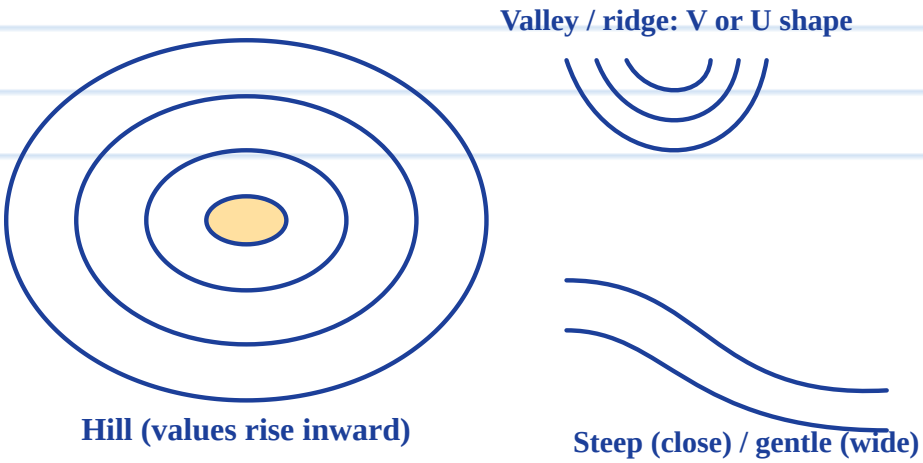


Fig 2.3 Contour patterns – hill, valley, slope

> Contour line: an imaginary line joining points of equal elevation on the ground; contour interval (CI) = vertical distance between successive contours (0.5, 1, 2, 5, 10 m depending on scale & terrain); horizontal equivalent (HE) = horizontal distance between them.

- Contour map = a plan showing relief by contour lines.

❖Characteristics of contours

- Contour lines are closed curves (may close outside the map); do not cross each other (except overhanging cliffs/ caves) and do not merge/ split (except vertical cliff).
- Close spacing → steep slope; wide spacing → gentle slope; equally spaced → uniform slope; straight parallel → plane slope.
- Concentric closed contours with higher values inside → hill; lower values inside → depression/ pond.
- V-shaped / U-shaped contours – pointing to higher ground = valley; pointing toward lower = ridge/ spur.

- Contour is perpendicular to the direction of steepest slope (watershed/ drainage lines cut contours at right angles). Contours cross a ridge or valley line at right angles.
- Ridge line = watershed; valley line = thalweg.
- Gradient = CI / HE (e.g. 1 in n).

❖Methods of contouring

- Direct method: staff is moved till the required level reading is found; points located & plotted – accurate, slow (small areas).
- Indirect method: (a) Grid / squares method – area divided into squares (5–20 m); levels at corners → interpolate contours; (b) Cross-section method – roads/ canals; (c) Radial lines method (tacheometric – hills, from a central station).
- Interpolation of contours: by estimation, arithmetic calculation, or graphical method (tracing paper/ parallel lines).

❖Uses of contour maps

- Selecting site for roads, railways, canals, dams & reservoirs (catchment area, storage capacity), pipelines, airports.
- Finding intervisibility between points, longitudinal sections, earthwork volumes (cut/fill), determination of catchment area & watershed, gradients, planning drainage, military & landscape planning.
- Capacity of reservoir: $V = \Sigma\ (A_1 + A_2)/2 \times CI$ (or prismoidal formula) where A = area enclosed by contour.

❖Example

> Contours 100 m and 102 m are 40 m apart (HE) on plan: gradient = CI/HE = 2/40 = 1 in 20 = 5 %. Contours of 100 m and 105 m at 25 m: slope = 5/25 = 1 in 5 (steep).

9. Additional Short Notes

❖Plane table surveying (brief)

- Surveying + plotting at the same time in the field using a drawing board (plane table) on a tripod, alidade, spirit level, plumb fork, trough compass; methods – radiation, intersection, traversing, resection.

❖Traversing

- A series of connected survey lines; **closed traverse** (polygon) or **open traverse**; angles & lengths measured; **Bowditch rule** for closing error; **Gale's traverse table**.

❖Modern survey tools

- **Total station, GPS/ GNSS (NavIC in India), Drone (UAV) photogrammetry, LiDAR, GIS & remote sensing, Survey of India (Dehradun)** is the national mapping body.

10. More Practice Problems

❖Problem A – Reduce the levels (collimation method)

> Staff readings (m) taken at 20 m intervals with the level fixed at one setup: 0.585 (BS on BM, RL 150.000), 1.015, 1.640, 2.220, 1.915, 0.320 (FS). H.I. = 150.000 + 0.585 = 150.585.

RLs: 149.570, 148.945, 148.365, 148.670, **150.265**. Check: $\Sigma BS - \Sigma FS$ = 0.585 – 0.320 = 0.265 = 150.265 – 150.000 ✔.

❖Problem B – Slope of a line from contours

> Two points lie on contours 220 m and 226 m; plan distance 120 m. Slope = 6/120 = **1 in 20**; gradient = **5 %**; angle = $\tan^{-1}(0.05)$ = **2.86°**.

❖Problem C – Bearings of a closed traverse

> Traverse ABCD, FB of AB = 45°. Interior angles B = 110°, C = 85°, D = 80°. BB of AB (bearing of BA) = 225°. For a clockwise traverse $FB_{next} = BB_{prev}$ – interior angle (add 360° if negative): **FB BC = 225° – 110° = 115°**; BB of BC = 295°; **FB CD = 295° – 85° = 210°**; BB of CD = 30°; **FB DA = 30° – 80° = –50° + 360° = 310°**; BB of DA = 130°; check **FB AB = 130° – 85°** (angle A = 360° – 110° – 85° – 80° = 85°) = 45° ✔.

❖Problem D – Chain correction

> A field measured 600 m with a 30 m chain found 0.1 m too long (30.10 m). True distance = 600 × 30.10/30 = **602.0 m**.

❖Problem E – Curvature & refraction

> Sight length = 300 m = 0.3 km → correction = 0.0673×0.3^2 = 6.06 mm (reading is too high; subtract from staff reading).

11. Errors in Levelling

- **Instrumental:** collimation error (line of sight not parallel to bubble axis) – cancel by keeping **BS = FS distance**; staff graduation errors; bubble sluggish.
- **Personal:** staff not held vertical, wrong reading/ booking, parallax, eye fatigue (staff reading is **least when staff is tilted** → hold vertical & wave slowly to read minimum).
- **Natural:** curvature, refraction, wind, temperature, settlement of tripod.
- **Mistakes:** misreading the staff (e.g. 1.234 as 1.324), booking in wrong column, moving the instrument at a change point.
- Allowable closing error in levelling = $\pm C\sqrt{K}$ mm (K = distance in km): about $\pm 24\sqrt{K}$ for ordinary and $\pm 12\sqrt{K}$ for precise levelling.

12. Comparison Tables

Feature	Chain survey	Compass survey	Theodolite
Measures	Linear only	Bearings	H & V angles
Accuracy	Low	Medium (30′)	High (20″)
Use	Small open flat area	Rough traverse	Precise control, setting out
Local attraction	No	Yes	No
Prismatic vs surveyor's compass	Prismatic	Surveyor's	
Graduation	WCB (0–360°)	QB (0–90°)	
Card	Fixed to the box (needle free)	Fixed to the needle (rotates)	
Reading	Through prism, any distance	Direct through glass lid	

- Quick formulae to remember:
- BB = FB ± 180°; QB ↔ WCB table; True B = Magnetic B ± declination (E +).

- H.I. = RL + BS; RL = H.I. – FS; ΣBS – ΣFS = ΣRise – ΣFall = Last RL – First RL.
- Gradient = CI / HE; Sum of interior angles = (2n – 4) × 90°.

13. Important Questions

- Define surveying; state its objectives and principles.
- Explain chain surveying: instruments, ranging, offsets, errors.
- What is bearing? Distinguish WCB & QB; convert & solve problems on FB, BB, local attraction.
- Define levelling terms (BM, BS, IS, FS, HI, CP, RL). Explain dumpy level & its adjustments.
- Reduce the levels (rise & fall / HI) with check.
- Define contour; explain characteristics, methods & uses of contouring.

Practise at least 4–5 levelling table problems – guaranteed marks!

BASIC CIVIL & MECHANICAL ENGINEERING

Subject Code: 23A01201T | UNIT – III : TRANSPORTATION, WATER RESOURCES & ENVIRONMENTAL ENGINEERING

I. Importance of Transportation in a Nation's Economic Development

> **Transportation** = movement of people and goods from one place to another. It is the **lifeline of a nation's economy.**

- **Modes:** roads (highways), railways, waterways (inland, sea), airways, pipelines, ropeways.

❖Importance

- Links **production centres, markets, raw material sources** → trade & industrial growth.
- Provides **employment** (drivers, rail/ port staff, construction) and promotes tourism.
- **Integrates** the country socially & politically; access to health, education, remote & border areas.
- Reduces **cost of goods**, transport cost, time; gives **price stability**, balanced regional development (agriculture → mandi/ export).
- Essential for **defence**, disaster relief, urban development, GDP growth (transport ≈ 6% of GDP).
- Indian scenario: road network ≈ **6.3 million km** (2nd largest in the world); National Highways ≈ 1.46 lakh km (Bharatmala, Golden Quadrilateral, NHAI); railways ≈ 68,000 route km (Indian Railways – 4th largest); 13 major + 200 minor ports (Sagarmala); 150+ airports (UDAN); metro rail in 20+ cities.

❖Road classification (Nagpur Plan)

- National Highways (NH), State Highways (SH), Major District Roads (MDR), Other District Roads (ODR), Village Roads (VR); also expressways.

❖Components of a highway

- Carriageway, shoulders, median, kerb, side drains, camber (1 in 50), embankment, pavement, culverts/ bridges, road signs & markings.

2. Types of Highway Pavements

> Pavement = durable surface material laid over the subgrade to carry traffic loads safely and distribute them to the soil beneath. Two types: Flexible & Rigid (and composite).

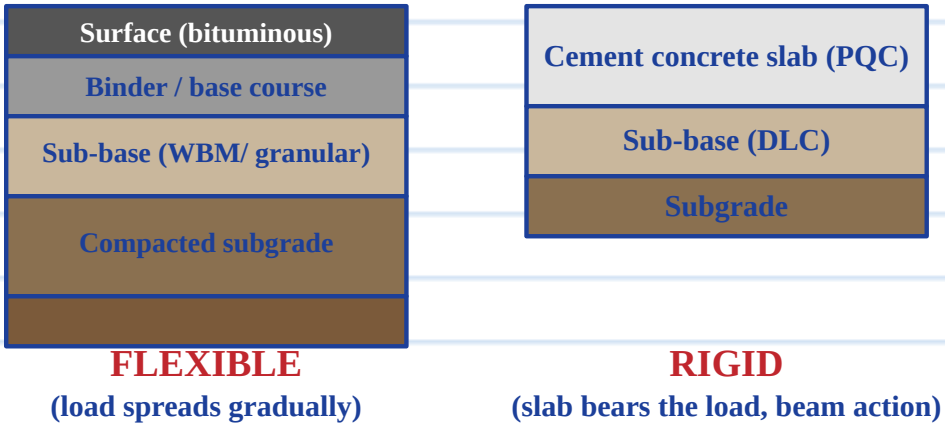


Fig 3.1 Flexible vs rigid pavement composition

❖2.1 Flexible Pavements

- Made of **bituminous (asphalt) surface** over granular layers; **load is transferred by grain-to-grain contact**, spreading in a **cone shape** and decreasing with depth; each layer shows **small deflection** under wheel load (flexes).
- Layers (top → bottom):
  - **Surface/ wearing course** – bituminous concrete (BC), semi-dense BC, mastic asphalt – resists wear, waterproof.
  - **Binder course** – dense bituminous macadam (DBM).
  - **Base course** – WBM (water bound macadam), WMM (wet mix macadam).
  - **Sub-base course** – granular (GSB, moorum, gravel).
  - **Subgrade** – compacted natural soil (CBR-based design).
- Design method: **CBR method (IRC 37)**, group-index method.
- Life: 10–15 years; **initial cost low**, maintenance high (potholes, rutting, bleeding).

❖2.2 Rigid Pavements

- Cement concrete slab (PQC – pavement quality concrete M30–M40, thickness 200–300 mm) over a dry lean concrete (DLC) sub-base; the stiff slab acts as a beam, spreading load over a large area of subgrade (beam action).
- Joints (expansion, contraction, longitudinal, construction) with dowel bars & tie bars to control cracking.
- Types: Plain concrete (JPCP), Jointed reinforced (JRCP), Continuously reinforced (CRCP), Prestressed.
- Design: Westergaard's theory, IRC 58. Life: 30–40 years; initial cost high, maintenance low.

❖Simple differences – Flexible vs Rigid

Point	Flexible	Rigid
Surface	Bituminous	Cement concrete slab
Load transfer	Grain to grain, layer by layer	Slab bending (beam action)
Design	Thickness by CBR	Slab thickness by flexural strength
Deformation	Takes deformation (settlement)	Rigid, may crack
Joints	No joints	Joints required
Initial cost	Low	High
Maintenance	High, frequent	Low
Life	10–15 yr	30–40 yr
Night visibility	Poor (dark)	Good (light colour)
Fuel/ temp	Affected by heat (melts)	Heat resistant
Repair	Easy, quick opening	Difficult
Stage construction	Possible	Not possible

3. Basics of Harbour Engineering

> A harbour is a sheltered area of water where ships can anchor/ berth safely for loading, unloading, repairs and refuge from storms.

- Classification: Natural (e.g., Mumbai, Kochi), Semi-natural (protected by break-water on one side), Artificial (fully protected by break-waters – Chennai); by use – Commercial, Refuge, Fishery, Naval, Military.

- **Components:** break-waters (protect from waves), jetties, wharves, quays, piers (berthing structures), dolphins, dredged channels, docks (wet dock with lock gate, dry dock for repair), navigation aids (lighthouse, buoys), transit sheds, warehouses, cranes, container yard, fenders (absorb ship impact).
- **Site selection:** sufficient water depth (≥ 10–15 m), natural protection, good anchorage, hinterland connectivity (rail/ road), low siltation, availability of land.
- **Terms:** Draft (depth of ship under water), tide, tidal range, wave action, siltation, dredging.
- **Major ports of India:** Mumbai, JNPT, Kandla (Deendayal), Mormugao, New Mangalore, Kochi, Tuticorin (V.O. Chidambaranar), Chennai, Ennore (Kamarajar), Visakhapatnam, Paradip, Kolkata (Haldia). Sagarmala project.

4. Basics of Tunnel Engineering

> **Tunnel** = an underground passage (horizontal/ slightly inclined) driven through hill/ soil/ under river without disturbing the surface; length > 100 m, length > width.

- **Purposes:** railways/ roads through mountains, metro, water conveyance (hydropower, irrigation), sewers, utilities, mines, defence.
- **Advantages:** shorter route, saves time & fuel, avoids traffic and land acquisition, protected from weather.

❖Classification

- By alignment (horizontal/ vertical/ inclined), by material (soil, hard rock), by position: hill tunnel, subaqueous (river), urban/ cut-and-cover tunnel.
- Shapes: circular, horse-shoe (D-shape), egg-shaped, rectangular.

❖Methods of tunnelling

- Drill & blast (hard rock) → mucking (removal) → support (steel ribs, shotcrete, rock bolts) → lining (concrete). NATM (New Austrian Tunnelling Method).
- TBM (Tunnel Boring Machine) – full-face mechanical excavation, used for metro/ soft ground.

- Cut and cover – open trench roofed over (shallow metro).
- Immersed tube for river crossings.
- Needs ventilation, drainage, lighting and safety.
- Examples in India: Atal Tunnel (Rohtang, 9.02 km, Himachal), Pir Panjal rail tunnel (J&K), Konkan Railway tunnels, Delhi Metro, Chenani–Nashri tunnel, Banihal.

5. Basics of Airport Engineering

> Airport = a facility where aircraft take off, land, are parked, serviced, and passengers/ cargo are handled.

- Components: Runway (take-off/ landing strip, oriented along prevailing wind; length 1000–4000 m), taxiway (connects runway & apron), apron (parking & loading area), terminal building, control tower (ATC), hangars, navigation aids (ILS, VOR), lighting, windsock, parking, cargo & fuel stations.
- Site selection: away from city (noise) but accessible; flat, well-drained land; clear approach zones (no obstructions); good soil; wind direction; meteorology (fog); room for expansion.
- Runway orientation: along prevailing wind (wind rose, wind coverage ≥ 95%); aircraft takes off/ lands against wind.
- Runway length depends on aircraft type, altitude (↑ height → ↑ length), temperature, gradient, wind: correction for elevation, temperature, gradient.
- Classification (ICAO): Code A–F by wingspan (A: < 15 m... F: 65–80 m); International, domestic; Greenfield airports (Navi Mumbai, Noida, Bhogapuram), regional – UDAN scheme.
- Pavements: flexible (asphalt), rigid (concrete, heavy aircraft).
- Zones: approach, turning, transition, horizontal and conical surfaces (imaginary surfaces for obstruction clearance).

6. Basics of Railway Engineering

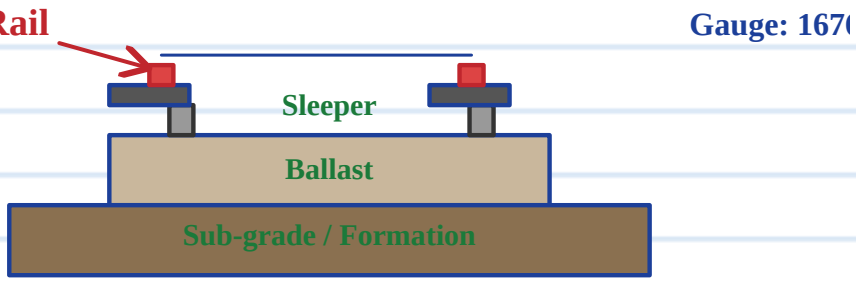


Fig 3.2 Railway track cross-section (Indian Broad Gauge = 1676 mm)

> **Railway** = guided transport on a **permanent way (track)** with steel rails, carrying trains hauled by locomotives; cheapest for **heavy and long-distance** traffic.

❖Components of a track (permanent way)

- **Rails:** steel (52 kg/m, 60 kg/m, 90 UTS), flat-footed type; carry loads & guide wheels.
- **Sleepers:** transverse supports under rails – **concrete (PSC)** most common, steel, wooden, cast-iron; spacing ~ 60 cm (1660 per km). They hold gauge, distribute load, insulate.
- **Fasteners:** fish plates/ elastic clips/ pandrol clips, **bolts, spikes, base plates.**
- **Ballast:** crushed stone (20–50 mm) 250–350 mm thick under sleepers – drains, resilient, distributes load, holds track in position.
- **Formation/ Subgrade** and embankment/ cutting.

❖Gauge (distance between inner faces of rails)

Gauge	Width	Where
Broad Gauge (BG)	1676 mm	Main Indian lines (~ 96%)
Metre gauge (MG)	1000 mm	Few branch lines
Narrow gauge (NG)	762 / 610 mm	Hill railways (Darjeeling, Nilgiri, Kalka-Shimla)
Standard gauge	1435 mm	Metro, world, high-speed

- **Other topics: points & crossings (turnouts)** to shift trains from one track to another; **signals, interlocking;** stations & yards; **gradient, super-elevation (cant)** on curves; **coning of wheels (1 in 20); creep.**

- Modern: Electrification (25 kV AC), Vande Bharat, Dedicated Freight Corridors (DFC), Metro, Mono, bullet train (MAHSR).

7. WATER RESOURCES & ENVIRONMENTAL ENGINEERING – Introduction

- Water covers 71% of earth but only 2.5% is fresh, and < 1% is easily accessible (lakes, rivers, groundwater).
- India: 16% of world population, 4% of water resources; average annual rainfall 1170 mm; monsoon (June–Sept) gives 75% of rain → need storage & management.
- Water resources engineering – assessment, planning, development, utilisation & control of water for drinking, irrigation, power, industry, navigation, flood control.
- Environmental engineering – provides safe water, sanitation, pollution control.
- Water demand (urban): 135 LPCD (litres per capita per day) for domestic (CPHEEO), 40–70 LPCD in rural; includes domestic, industrial, public, fire, losses.

8. Sources of Water

- Surface sources: rivers, streams, lakes, ponds, reservoirs (impounding), canals – turbid, bacteria, easily polluted, large quantity, needs treatment.
- Ground water sources: wells (open/ dug, tube wells, bore wells), springs, infiltration galleries – clear, low turbidity, hard (more minerals), usually safe, limited yield; recharge by rain.
- Rain water: collected from roofs; pure but acidic, needs storage.
- Others: sea water (desalination), recycled waste-water.

❖Surface vs underground source

Feature	Surface	Ground
Turbidity	High	Low
Bacteria	More	Less
Hardness	Soft	Hard
Quantity	Large	Limited
Treatment	Full (coagulation, filtration)	Usually disinfection only

9. Quality of Water & Specifications

> Potable water = water that is safe to drink – clear, colourless, odourless, tasteless, free from pathogens, toxic substances, and palatable. Standard: IS 10500 : 2012 (BIS drinking water specification); also WHO guidelines.

❖Impurities

- Suspended (silt, clay, algae), colloidal, dissolved (salts, gases, organics), bacteria/ viruses/ protozoa (cholera, typhoid, hepatitis, diarrhoea).

❖Water quality parameters

- Physical: turbidity (NTU), colour (Hazen units), taste & odour, temperature, total solids/ TDS, conductivity.
- Chemical: pH, hardness (Ca, Mg – $CaCO_3$ mg/L), alkalinity, chlorides, sulphates, fluoride, nitrates, iron, manganese, arsenic, heavy metals, dissolved oxygen (DO), BOD, COD.
- Biological/ bacteriological: coliform bacteria (E. coli) – indicator; MPN/ 100 mL.

❖Drinking-water limits (IS 10500, desirable / permissible)

Parameter	Acceptable	Permissible (no alternate)
pH	6.5 – 8.5	no relaxation
Turbidity	1 NTU	5 NTU
TDS	500 mg/L	2000 mg/L
Total hardness ($CaCO_3$)	200 mg/L	600 mg/L
Chloride	250 mg/L	1000 mg/L
Sulphate	200 mg/L	400 mg/L
Fluoride	1.0 mg/L	1.5 mg/L
Nitrate	45 mg/L	no relaxation
Iron	0.3 mg/L	no relaxation (1.0)
Arsenic	0.01 mg/L	0.05 mg/L
Residual chlorine	0.2 mg/L	1.0 mg/L
Coliform / E. coli	Nil / 100 mL	Nil

- Effects: excess fluoride → fluorosis; nitrate → blue-baby disease; arsenic → cancers; hardness → scale & soap waste.

❖Water treatment (brief flow)

Raw water → Screening → Sedimentation (+ coagulation: alum) → Filtration (rapid/ slow sand) → Disinfection (chlorination, UV, ozone) → Storage → Distribution

- Aeration removes gases/ iron; softening removes hardness (lime-soda, ion exchange); RO for TDS; disinfection with chlorine (0.2–0.5 mg/L residual).
- Waste-water (sewage) treatment: preliminary (screens, grit) → primary (sedimentation) → secondary (biological – activated sludge, trickling filter) → tertiary → disinfection; sludge digestion.

10. Introduction to Hydrology

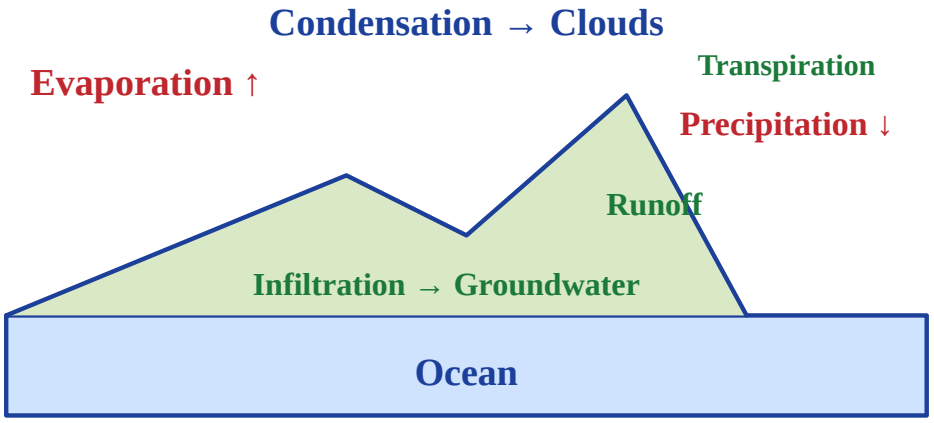


Fig 3.3 Hydrologic cycle

> Hydrology is the science dealing with the occurrence, circulation, distribution and properties of water on earth – above, on and below the surface.

❖Hydrologic cycle

- Evaporation (sun heats oceans/ lakes) + transpiration (plants) = evapotranspiration → condensation (clouds) → precipitation (rain, snow, hail, dew) → interception, infiltration (into soil), percolation (to ground water), surface runoff (streams) → rivers → ocean.
- Water balance equation: $P = R + E + I + \Delta S$ (Precipitation = Runoff + Evapotranspiration + Infiltration + change in storage).

❖Precipitation

- Forms: rainfall, snow, sleet, hail, drizzle, dew, frost.

- Types of rainfall: convective (heating of ground – tropical), orographic (air forced up mountains – Western Ghats, North-east), cyclonic (low-pressure; frontal & non-frontal – coastal A.P., Odisha).
- Measurement: rain gauge – non-recording (Symon's gauge, 127 mm/ 5" funnel) and recording (tipping-bucket, float, weighing); unit mm; rain gauge density 1 per 520 km^2 (plain).
- Average rainfall over a catchment: arithmetic mean, Thiessen polygon method, isohyetal method.

Arithmetic mean = $(P_1+P_2+...+P_n)/n$; Thiessen mean = $\Sigma(P_iA_i)/\Sigma A_i$

❖Runoff

- Runoff = part of rainfall that flows over land to streams. Depends on rainfall intensity/ duration, catchment area, slope, soil, land use, vegetation, urbanisation.
- Rational formula (peak discharge): $Q = (1/360)\ C\ i\ A$ (Q in m^3/s, i in mm/h, A in hectares, C = runoff coefficient 0.1 – 0.95).
- Hydrograph (discharge vs time), catchment/ watershed, stream gauging (current meter), infiltration indices (φ-index).
- Flood, drought, base flow.

❖Example

Catchment A = 50 ha, C = 0.4, i = 60 mm/h → $Q = (1/360) \times 0.4 \times 60 \times 50 = 3.33\ m^3/s$.

Gauges P_1 = 80, P_2 = 100, P_3 = 120 mm → arithmetic mean = 100 mm.

II. Rainwater Harvesting (RWH)

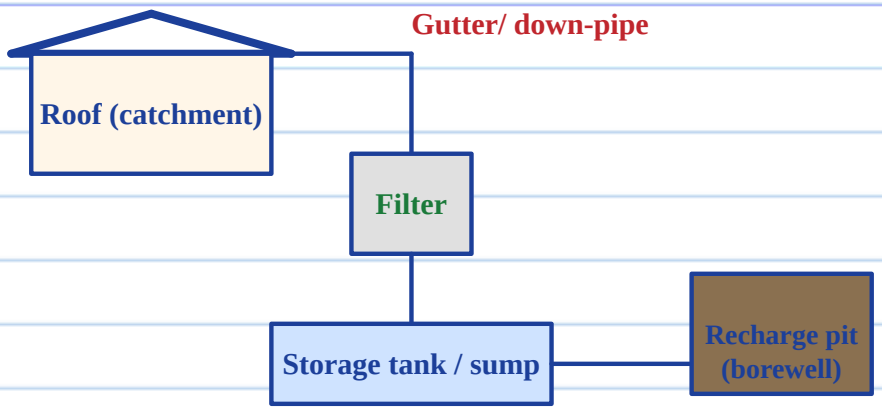


Fig 3.5 Rooftop rainwater harvesting system

> Rainwater harvesting = collecting, storing and using rain water (or recharging groundwater) where it falls, instead of letting it run off.

❖Need

- Falling water tables (over-exploitation), shortage in urban areas, floods & runoff, poor groundwater quality, increases self-sufficiency & saves energy of pumping.

❖Methods

- (a) Rooftop harvesting – storage: roof water → gutters → down pipes → first-flush diverter → filter (sand/ gravel/ charcoal) → underground sump/ overhead tank for direct use.
- (b) Recharge methods: recharge pit/ trench, recharge shaft, recharge through borewell/ dug well, percolation tank, check dams, gully plugs, contour bunds, farm ponds; injection wells for deep aquifers.
- Traditional systems: johads, kunds, baoli/ stepwells, tankas (Rajasthan), ahar-pynes, eris (Tamil Nadu), katas, bhandaras, cheruvu (A.P.).

❖Components of a roof-top system

- Catchment (roof), gutters & conduits, first-flush device (discard first 1 mm of rain), filter unit, storage tank or recharge structure, overflow.
- Design: Annual harvestable volume $V = A \times R \times C$ (A = roof area, R = rainfall, C = coefficient ≈ 0.8–0.85).

❖Example

> Roof 100 m^2, annual rain 800 mm = 0.8 m, C = 0.8: $V = 100 \times 0.8 \times 0.8 = 64\ m^3 = 64{,}000$ L/year.

❖Benefits

- Increases ground water table, reduces runoff/ flooding, improves water quality (dilutes fluoride/ salts), cheap, easy to maintain, reduces dependence on piped supply.
- Made mandatory in many cities (A.P. – "Neeru-Chettu", Hyderabad Water Board, Chennai).

12. Water Storage & Conveyance Structures

❖Storage structures

- Dams & reservoirs, tanks (ponds), check dams, storage tanks (ground-level, elevated, underground sumps), service reservoirs, balancing reservoirs, water towers.
- Reservoir zones: Dead storage (below minimum level, silt) → Live/ useful storage (between MDDL & FRL) → Flood/ surcharge storage (above FRL up to MWL).

❖Conveyance structures

- Canals (lined/ unlined), pipelines (CI, DI, PVC, HDPE, RCC, steel), aqueducts, siphons, flumes, tunnels, culverts, pumps with rising mains, penstocks.
- Pipe flow – Manning/ Hazen–Williams formulas. $Q = A\,V$.
- Distribution systems: gravity, pumping, dual; layouts – grid-iron, ring, radial, dead-end; valves, hydrants, meters, water towers maintaining 7–15 m residual head.

13. Dams and Reservoirs – Simple Introduction

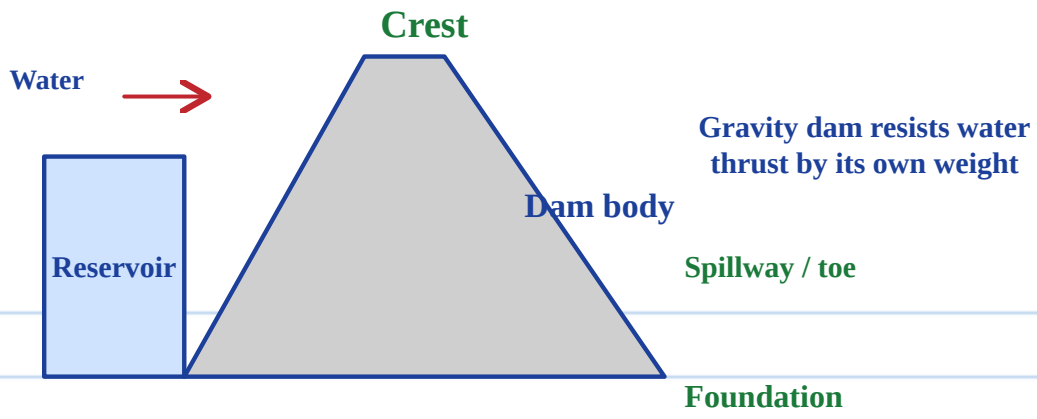


Fig 3.4 Gravity dam (profile) retaining a reservoir

A dam is a barrier constructed across a river to store water, forming a reservoir. India has ~5,700 large dams (third in world). Largest: Tehri (261 m, Uttarakhand), Bhakra (226 m), Hirakud, Nagarjuna Sagar (A.P.–Telangana, masonry), Srisailam, Polavaram, Sardar Sarovar, Krishnaraja Sagar.

❖Purposes

- Irrigation, hydroelectric power, drinking & industrial water supply, flood control, navigation, fisheries, recreation, groundwater recharge.

❖Classification

- By function: storage, diversion, detention, multipurpose.
- By material: masonry & concrete gravity, arch, buttress, earth-fill (embankment), rock-fill, steel/ timber.
- By structure/ hydraulic design: overflow (spillway) & non-overflow.
- Gravity dam: resists water thrust by own weight; needs good rock foundation; trapezoidal section; made of concrete/ masonry.
- Arch dam: curved in plan; transfers load to abutments; for narrow gorges (Idukki).
- Earth dam: local soil, impervious core, flatter slopes; for weak foundations; cheaper.

❖Components

- Crest, upstream/ downstream face, heel & toe, foundation, galleries, spillway (passes surplus flood safely), gates, outlets/ sluices, energy dissipators (stilling basin), fish ladder, power house (hydro dams).

❖Forces on a gravity dam

- Self weight (W), water pressure ($P = \frac{1}{2} \gamma_w H^2$ per metre), uplift pressure, silt pressure, wave, ice, earthquake, wind forces.
- Stability: safe against overturning, sliding, tension & crushing (no tension in base).

❖Reservoir

- Planning: capacity – elevation curve (from contour map), site selection (narrow gorge, wide valley, good rock, large catchment, low evaporation), sedimentation, submergence & rehabilitation (R & R).
- Levels: MDDL, FRL (full reservoir level), MWL (maximum water level); freeboard.

❖Advantages & problems

- Benefits: water, power, flood control. Problems: displacement, submergence of forests, siltation, seismicity, health (water-borne diseases), ecological downstream effects, high cost.

14. Solved Problems

❖Problem 1 – Water demand

> Population 20,000 × 135 LPCD = 2,700,000 L/day = **2700 m^3/day** = 31.25 L/s.

❖Problem 2 – Hardness

> Ca = 60 mg/L, Mg = 24 mg/L: hardness (as $CaCO_3$) = 2.5 Ca + 4.1 Mg = 150 + 98.4 = **248 mg/L** (hard; above 200 desirable).

❖Problem 3 – Hydrostatic force on dam

> Water depth H = 40 m: $P = ½ \times 9.81 \times 40^2$ = **7848 kN per metre**, acting at H/3 = **13.33 m** from base.

❖Problem 4 – Reservoir capacity

> Areas enclosed by contours at 100, 102, 104 m = 0.5, 1.2, 2.0 km^2 → V = (0.5+1.2)/2×2 + (1.2+2.0)/2×2 = 1.7 + 3.2 = **4.9 million m^3**.

15. Quick Revision & Important Questions

Topic	Key point
Highway pavements	Flexible = bitumen, grain-to-grain; Rigid = concrete slab, beam action
Railway	BG = 1676 mm; rails, sleepers, ballast
Water quality	pH 6.5–8.5; coliform nil; TDS ≤ 500 mg/L
Hydrology	$P = R + E + I + \Delta S$; $Q = CiA/360$
RWH	$V = A \times R \times C$
Dam	Gravity dam – self weight; crest, spillway, reservoir

- Explain the importance of transportation in economic development.
- Differentiate flexible and rigid pavements with sketches.
- Write short notes on harbour, tunnel, airport and railway engineering.
- Sources of water; quality parameters & drinking-water specifications.
- Hydrologic cycle; rainwater harvesting methods; dams and reservoirs.

BASIC MECHANICAL ENGINEERING

Subject Code: 23A01201T (Part B) | UNIT – I : INTRODUCTION TO MECHANICAL ENGINEERING & ENGINEERING MATERIALS

I. Introduction – Role of Mechanical Engineering

> Mechanical Engineering is the branch of engineering that applies the principles of physics, mathematics and materials science to design, analyse, manufacture and maintain mechanical systems – machines, engines, tools, vehicles and energy systems.

- Called the "mother of all engineering branches" – broadest, oldest after civil; deals with motion, force, energy, heat and materials.
- Core areas: (i) Thermal engineering (heat engines, boilers, refrigeration), (ii) Design engineering (machine elements, CAD), (iii) Manufacturing/ production engineering, (iv) Fluid mechanics & machinery, (v) Mechatronics & robotics, (vi) Materials engineering.

❖Role in industries

- Designs & maintains machine tools, boilers, turbines, pumps, compressors, conveyors, presses.
- Production: converts raw material to finished goods – cars, appliances, electronics casings, tools.
- Plant engineering: installation, maintenance, energy audit, quality control, automation.
- Supports sectors such as power, steel, cement, textile, chemical, food processing, mining, defence.

❖Role in society

- Transport (cars, trains, ships, aircraft), energy (power plants, wind & solar machines), healthcare equipment (ventilators, prosthetics, MRI machines), air-conditioning & refrigeration (food/ medicine preservation), agriculture machinery (tractors, pumps), household appliances, sports gear.
- Improves quality of life, productivity, safety; creates jobs and drives economic growth ("Make in India").

- Responsible for sustainability – efficient engines, renewable energy devices, EVs, recycling.

> Mechanical engineer = designs the machines that move the world

2. Technologies in Different Sectors

❖(a) Energy sector

- Thermal power plants (steam turbines, boilers), gas turbines & combined cycle, hydro turbines (Pelton, Francis, Kaplan), nuclear plants.
- Renewables: wind turbines, solar thermal collectors, biomass/ biogas, geothermal; energy storage; hydrogen & fuel cells.
- Energy efficiency: heat exchangers, waste-heat recovery, cogeneration, HVAC.

❖(b) Manufacturing sector

- Casting, forging, machining, welding; CNC machines, robots, automation, CAD/CAM; 3D printing (additive manufacturing); Industry 4.0 – IoT, AI, smart factories; lean manufacturing, quality (Six Sigma).

❖(c) Automotive sector

- IC engines (petrol, diesel), transmission (gears, clutch), brakes, suspension, steering, turbochargers, fuel injection (CRDI, GDI), emission control (catalytic converter, BS-VI), Electric vehicles (EV) & hybrids, ADAS, lightweight materials (Al, composites).

❖(d) Aerospace sector

- Aircraft & rockets: aerodynamics, jet engines (turbojet, turbofan), rocket propulsion, light & strong materials (titanium, aluminium alloys, carbon fibre composites), avionics, satellite launch (ISRO – PSLV, GSLV, Chandrayaan, Gaganyaan), drones/ UAV.

❖(e) Marine sector

- Ship design & propulsion – marine diesel engines, gas turbines, propellers, rudders, hull design (naval architecture), submarines, offshore platforms, ballast & pumps, corrosion protection, LNG carriers, green shipping (hybrid/ electric ferries).

❖(f) Others

- Biomedical (implants, prosthetics), Robotics, Construction machinery, HVAC, Railways (locomotives, coaches), Defence (tanks, missiles).

3. Engineering Materials – Introduction

> Engineering materials are the substances used to make machine parts, tools and structures. Selection depends on strength, hardness, ductility, toughness, density, thermal & electrical conductivity, corrosion resistance, machinability, cost, availability.

❖Classification

Engineering materials → Metals (ferrous, non-ferrous) / Ceramics / Polymers / Composites / Smart / advanced materials

❖Mechanical properties (quick recap)

- Strength – resistance to failure under load. Elasticity – regain shape after load removal. Plasticity – permanent deformation. Ductility – drawn into wires (mild steel, copper). Malleability – hammered into sheets (gold, lead). Brittleness – fracture without deformation (cast iron, glass). Toughness – energy absorbed before fracture. Hardness – resistance to indentation/ scratch. Fatigue – failure under repeated loads. Creep – slow deformation at high T. Stiffness (Young's modulus E).

4. Metals

> Metals are crystalline, with metallic bonding, good electrical & thermal conductivity, ductility, lustre, strength.

❖4.1 Ferrous metals (iron-based)

- Contain iron as the main constituent. Iron is obtained from ores (hematite Fe_2O_3, magnetite Fe_3O_4) in a blast furnace → pig iron (3.5–4.5% C).

Material	Carbon %	Properties	Uses
Wrought iron	< 0.08	Soft, ductile, tough, rust-resistant	Chains, hooks, railings
Mild steel	0.15–0.3	Ductile, weldable, strong	Bolts, shafts, structures
Medium carbon steel	0.3–0.6	Stronger, tougher	Gears, axles, rails
High carbon steel	0.6–1.5	Very hard, brittle	Cutting tools, springs, dies
Cast iron (grey)	2–4	Brittle, good casting, damping, compressive strength	Engine blocks, machine beds, pipes
White CI / Malleable CI / SG (ductile) iron	2–4	Hard / tougher	Crankshafts, pipes

- Alloy steels: steel + Cr, Ni, Mo, V, Mn, W: stainless steel (18% Cr + 8% Ni – corrosion-proof, utensils, surgical tools), HSS (high speed steel – W, Cr, V) for cutting tools, spring steel, tool steel.
- Heat treatment (annealing, normalising, hardening, tempering, case hardening) changes properties.

❖4.2 Non-ferrous metals (no/ little iron)

Metal / Alloy	Properties	Uses
Aluminium (Al)	Light (2.7 g/cc), corrosion-resistant, good conductor, ductile	Aircraft, cans, cables, engine pistons
Copper (Cu)	Excellent electrical & thermal conductor, ductile	Wires, heat exchangers
Brass (Cu + 30–40% Zn)	Strong, corrosion-resistant, machinable	Valves, fittings, musical instruments
Bronze (Cu + Sn)	Hard, wear resistant	Bearings, bushes, gears, statues
Zinc (Zn)	Low melting, corrosion-protecting	Galvanising, die-castings
Lead (Pb)	Heavy, soft, corrosion-proof	Batteries, radiation shield
Tin (Sn)	Soft, non-toxic	Solder, tin plating
Titanium (Ti)	High strength/ weight, corrosion-proof	Aerospace, implants
Duralumin (Al+Cu+Mg)	Strong & light	Aircraft frames
Magnesium (Mg)	Lightest structural metal	Auto & aero parts
Nickel alloys (Inconel, Monel)	High-temp strength	Gas turbine blades

- Advantages of non-ferrous: light, corrosion-resistant, good conductivity, easy fabrication, non-magnetic. Disadvantages: costly, lower strength than steel.

❖Ferrous vs non-ferrous

Feature	Ferrous	Non-ferrous
Main element	Iron	Other than iron
Magnetic	Yes (mostly)	No
Corrosion	Rusts (except stainless)	Resistant
Density	High	Usually low
Cost	Cheaper	Costlier
Strength	High	Moderate

5. Ceramics

> Ceramics are inorganic, non-metallic solids made of metallic + non-metallic elements (oxides, carbides, nitrides) by shaping and high-temperature firing (sintering); ionic/ covalent bonds.

- Properties: very hard, high compressive strength, high melting point (> 2000 °C), excellent wear & corrosion resistance, electrical & thermal insulators, low density; but brittle (poor tensile strength, thermal shock).
- Types: traditional – bricks, tiles, pottery, porcelain, cement, glass; advanced/ engineering – alumina (Al_2O_3), silicon carbide (SiC), silicon nitride (Si_3N_4), zirconia (ZrO_2), tungsten carbide.
- Uses: cutting tool inserts, abrasives (grinding wheels), furnace linings/ refractories, spark plug insulators, engine/ turbine parts (thermal barrier coatings), bio-ceramics (hip joints, dental), electronic substrates, capacitors, bullet-proof armour, space shuttle tiles.

6. Polymers (brief – to complete the classification)

- Long-chain organic molecules – plastics (thermoplastics soften on heating: PVC, polyethylene, nylon, ABS; thermosets harden permanently: Bakelite, epoxy) and elastomers (rubber).
- Light, corrosion-proof, insulating, easily moulded, cheap; but low strength & temperature limit. Uses: pipes, gears (nylon), bearings, insulation, bumpers.

7. Composites

> A composite material is made by combining two or more distinct materials (matrix + reinforcement) so that the result has properties better than each constituent, with a clear interface between them.

- Matrix (binder): polymer, metal or ceramic – holds fibres, transfers load, protects them.
- Reinforcement: fibres (glass, carbon, aramid/ Kevlar), particles, flakes.

❖Classification

- By matrix: PMC (polymer matrix – most common, e.g., FRP, CFRP), MMC (metal matrix – Al-SiC), CMC (ceramic matrix – SiC-SiC).

- By reinforcement: fibre-reinforced (continuous/ short), particle-reinforced (concrete, cermet), laminar/ sandwich (plywood, honeycomb panels).

❖Properties

- High strength-to-weight and stiffness-to-weight, corrosion resistance, fatigue resistance, tailor-made (anisotropic), design flexibility, low thermal expansion.
- Drawbacks: costly, difficult to repair/ recycle, anisotropic, delamination, brittle failure.

❖Examples & applications

- Fibre glass (GFRP) – boat hulls, tanks, pipes, bathtubs, wind-turbine blades.
- Carbon fibre (CFRP) – aircraft (Boeing 787, 50% composite), F1 cars, bicycles, sports goods, satellites, prosthetics.
- Reinforced concrete, plywood, fibre cement, Kevlar (bullet-proof vests), brake pads, tungsten carbide cermet tools.

> Concrete + steel = RCC; Glass fibre + resin = GFRP; Carbon fibre + epoxy = CFRP

8. Smart Materials

> Smart (intelligent) materials are materials whose properties (shape, stiffness, colour, conductivity) change in a controlled & reversible way in response to an external stimulus such as temperature, stress, electric/ magnetic field, light, pH.

❖Types, behaviour & uses

Smart material	Stimulus → Response	Applications
Shape memory alloys (SMA) – Nitinol (Ni-Ti)	Heat → returns to original shape	Stents, eyeglass frames, actuators, aircraft wing flaps, couplings
Piezoelectric – quartz, PZT	Stress ⇄ electric voltage	Sensors, lighters, ultrasound, microphones, energy harvesting, inkjet printers
Magnetostrictive – Terfenol-D	Magnetic field → change in length	Sonar, actuators
Electro-/ magneto-rheological fluids (ER/ MR)	Field → viscosity jumps	Adaptive dampers, shock absorbers, clutches
Thermochromic / photochromic	Temp/ light → colour change	Mood rings, smart glass, spectacles
Electrochromic	Voltage → opacity	Smart windows, aircraft windows
Self-healing polymers/ concrete	Crack → heals	Coatings, infrastructure
Hydrogels, fibre-optics, shape-memory polymers	pH/ temp → swell	Drug delivery, sensors

- **Advantages:** act as sensor + actuator, adaptive, light, compact, reduce parts. **Limitations:** high cost, fatigue life, complex control.
- Used in robotics, aerospace (morphing wings), biomedical, civil (smart structures), automotive (adaptive suspension).

9. Material Selection – Summary Table

Need	Best choice
Cheap, strong structure	Mild steel
Engine block / machine bed	Grey cast iron
Light & corrosion-proof	Aluminium alloy
Electrical wire	Copper
Cutting tools	HSS, tungsten carbide, ceramics
Aircraft high strength/ weight	CFRP, titanium
Surgical/ kitchen	Stainless steel
High temperature	Ceramics, Inconel
Sensors/ actuators	Piezo, SMA

10. Important Questions

- Define mechanical engineering; explain its role in industries and society.
- Mention the technologies used in energy, automotive, aerospace, marine sectors.
- Classify engineering materials. Differentiate ferrous & non-ferrous metals with examples.
- Write about ceramics, composites and smart materials with applications.
- Compare mild steel, cast iron, brass and bronze (composition & uses).

> Remember: Mild steel 0.15–0.3% C; High carbon steel 0.6–1.5% C; Brass = Cu + Zn; Bronze = Cu + Sn

11. Extra Facts for Quick Revision

- **Steel** = iron + carbon (< 2% C); **cast iron** = iron + carbon (> 2% C).
- **Heat treatment:** *Annealing* – soften & relieve stress (slow furnace cooling); *Normalising* – refine grain (air cooling); *Hardening* – heat above critical temp, quench in water/ oil; *Tempering* – reheat after hardening to reduce brittleness; *Case hardening* – hard surface, tough core (gears).
- **Alloy** = mixture of a metal with other elements to improve properties (e.g., steel, brass, bronze, solder = Pb + Sn).
- **Corrosion** – electrochemical deterioration (rusting: $Fe + O_2 + H_2O \rightarrow$ hydrated iron oxide). **Prevention:** painting, **galvanising (Zn coat)**, plating, alloying (stainless), cathodic protection, oiling.
- **Density:** steel 7850; Al 2700; Cu 8960; Ti 4500; CFRP ~ 1600 kg/m^3.
- **Melting points:** Al 660 °C, Cu 1085 °C, Fe 1538 °C, W 3422 °C, Zn 420 °C, Pb 327 °C.
- **Stress–strain curve (mild steel):** proportional limit → elastic limit → yield point → strain hardening → ultimate stress → necking → fracture. Young's modulus **E = stress/ strain** (steel 200 GPa; Al 70 GPa; Cu 110 GPa).

12. Structure of Metals (short)

- Atoms are arranged in **crystal lattices**: **BCC** (body-centred cubic – α-iron, Cr, W; strong), **FCC** (face-centred cubic – γ-iron, Al, Cu, Ni; very ductile), **HCP** (hexagonal close packed – Zn, Mg, Ti; less ductile).
- **Grain size**: small grains → higher strength (Hall–Petch); heat treatment & cold working control grains.
- **Iron-carbon alloys phases**: ferrite (soft), **austenite** (non-magnetic, high T), **cementite** Fe_3C (very hard, brittle), **pearlite** (ferrite + cementite lamellae), **martensite** (very hard, from quenching).
- **Eutectoid steel** = 0.8% C (723 °C).

13. Testing of Materials (basics)

- **Tensile test** (UTM): specimen pulled until it breaks → gives yield strength, ultimate strength, % elongation, % reduction in area.
- **Hardness tests**: **Brinell** (HB – steel ball), **Rockwell** (HRC – diamond cone), **Vickers** (HV – diamond pyramid), **Mohs scale** (minerals).
- **Impact test**: **Charpy / Izod** – energy absorbed to break a notched bar (toughness).
- **Fatigue test** (rotating beam), **creep test**, **bend test**, **NDT** (ultrasonic, X-ray/ radiography, dye-penetrant, magnetic particle) for finding cracks without damage.

14. Materials in Real Products – Case Studies

Product / Part	Material	Why
Car body	Mild/ HSLA steel, Al	Strength, formability, weight
Engine block	Grey CI / Al alloy	Casting, damping / lightness
Piston	Aluminium alloy	Light, good conduction
Crankshaft, gears	Alloy steel, SG iron	Strength, fatigue resistance
Bearings	Bronze, babbitt	Low friction, wear
Radiator	Copper/ Al	Heat transfer
Tyres	Rubber + steel cord	Elasticity, strength
Aircraft skin	Al alloys, CFRP	Strength/ weight
Turbine blades	Ni superalloy + ceramic coat	High temperature
Ship hull	Mild steel, FRP (small boats)	Strength, cost
Cutting tool	HSS, carbide, ceramic	Hardness at high temperature
Electrical wire	Copper, Al	Conductivity
Medical implant	Ti, stainless steel, bioceramic	Bio-compatibility

15. Sector Technology Snapshot

- **Energy:** efficiency of a steam plant ≈ 35–40%, combined cycle gas plant ≈ 55–60%; hydro ≈ 90%; wind turbine $C_p \leq 0.59$.
- **Automotive:** typical petrol engine efficiency 25–30%, diesel 35–40%; EV motor ≈ 90%; BS-VI emission standard since 2020.
- **Aerospace:** lift = $\frac{1}{2} \rho V^2 S C_L$; thrust from Newton's 3rd law (jet/ rocket); ISRO's Mars/ Moon missions use cryogenic engines.
- **Marine:** buoyancy (Archimedes' principle) – ship floats when weight of water displaced = weight of ship; propellers efficiency 60–70%.
- **Manufacturing:** automation + robotics + AI → Industry 4.0; additive manufacturing reduces waste.

16. Key Definitions (for 2-mark answers)

- **Ferrous metal:** metal containing iron. **Non-ferrous:** without iron (Al, Cu, Zn...).
- **Alloy:** metal mixed with other elements. **Ceramic:** inorganic non-metallic, sintered.

- **Composite:** matrix + reinforcement. **Smart material:** responds to external stimulus.
- **Piezoelectric effect:** pressure ⇄ voltage. **Shape memory effect:** deformed material returns on heating.
- **Machinability:** ease of cutting. **Weldability:** ease of joining by welding. **Castability:** ease of casting.
- **Anisotropic:** properties differ with direction (composites). **Isotropic:** same in all directions (metals).

17. Short Quiz (Practice)

- Which metal is called the "lightest structural metal"? → **Magnesium.**
- Brass is an alloy of? → **Copper + Zinc.** Bronze? → **Copper + Tin.**
- Which cast iron is brittle but has good damping? → **Grey cast iron.**
- Stainless steel main alloying elements? → **Chromium (≥ 10.5%) + Nickel.**
- Which ceramic is used for spark plug insulators? → **Alumina.**
- Nitinol is an example of? → **Shape memory alloy.**
- CFRP stands for? → **Carbon fibre reinforced polymer.**
- Which property allows drawing into wires? → **Ductility.**

BASIC MECHANICAL ENGINEERING

Subject Code: 23A01201T (Part B) | UNIT – II : MANUFACTURING PROCESSES & THERMAL ENGINEERING

1. Manufacturing – Introduction

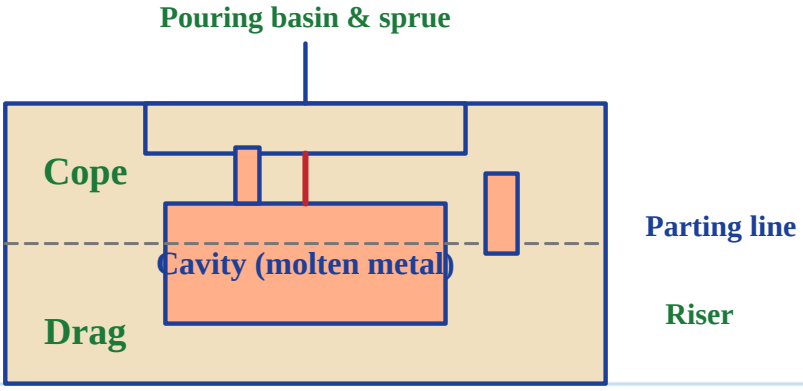
Manufacturing = converting **raw materials** into **finished useful products** through a sequence of processes (using machines, tools, labour, energy).

Manufacturing processes → Casting / Forming / Joining / Machining / Additive (3D printing)

- Selection depends on material, shape, size, quantity, accuracy, surface finish, cost.

2. Casting

Casting: pouring **molten metal** into a **mould cavity** of the required shape and allowing it to **solidify**; the solid piece is the **casting**.



Sand mould (two-part box)

Fig 1.1 Sand casting mould

❖Principle & steps

- **Pattern making** (wood/ metal/ wax/ plastic, with allowances: shrinkage, machining, draft, shake, distortion) → **mould making** (sand mould with gating system: **pouring basin**, **sprue**, **runner**, **gate**; **riser** to feed shrinkage; **cores** make hollow parts) → **melting** (cupola, induction/ electric arc furnace) → **pouring** → **solidification & cooling** → **shake-out/ fettling** (cleaning) → **inspection**.
- **Pattern** = replica of the part (made slightly larger); **mould** = cavity; **core** = sand shape forming internal cavities; **cope** (top box), **drag** (bottom box), **parting line**.

- **Mould sand:** silica sand + clay (bentonite) + water; needs permeability, strength, refractoriness.

❖Types of casting

- **Sand casting** – cheap, any size/ metal (engine blocks, pipes). **Die casting** (permanent metal die; pressure die – Al, Zn parts: carburettors, gear housings) – high precision, fast. **Investment (lost wax) casting** – intricate, high finish (turbine blades, jewellery). **Centrifugal casting** – pipes, liners. **Shell moulding, continuous casting** (steel billets).

❖Advantages / limitations

- Complex shapes, any size (few grams to tonnes), cheap for mass production, wide material choice. **Defects:** blow-holes, shrinkage cavities, cold shuts, misruns, hot tears, porosity; needs finishing; lower strength than forged.
- **Applications:** engine blocks, cylinder heads, pistons, pump casings, machine beds, bells, statues, pipes, wheels, valves.

3. Forming (Metal Working)

> **Forming** = **plastic deformation** of metal into the required shape by applying force, **without removing material** (no chips); volume remains constant.

- **Hot working** (above recrystallisation temperature – easy shaping, no strain hardening) vs **Cold working** (below – better strength & finish, closer tolerance, needs more force).

❖Main forming processes

- **Forging:** compressing heated metal with **hammers/ presses/ dies** – improves grain flow & strength (crankshafts, connecting rods, spanners, axes). Types: open die, closed (impression) die, drop, press, upset.
- **Rolling:** passing metal between **rotating rolls** to reduce thickness/ make shapes – plates, sheets, rails, I-beams, bars (hot & cold rolling).
- **Extrusion:** forcing metal through a **die** to get long constant section (pipes, rods, Al window frames, tubes). Direct/ indirect.
- **Drawing:** pulling metal through a die to reduce diameter – **wire drawing**, tube drawing, **deep drawing** (cups, utensils, car panels).
- **Sheet-metal operations:** shearing, blanking, punching, bending, stamping/ pressing, spinning.

- Other: powder metallurgy (compacting & sintering metal powders – gears, bearings, tungsten filaments), rolling of threads.

❖Comparison

Casting	Forming
Liquid metal	Solid metal deformed
Complex shapes	Simpler shapes
Lower strength	Higher strength (grain flow)
More defects	Good integrity

4. Joining Processes

> Joining = connecting two or more parts permanently or temporarily.

- Temporary (mechanical): bolts, nuts, screws, rivets (semi-permanent), keys, pins, cotters, couplings.
- Permanent: welding, brazing, soldering, adhesive bonding, riveting.

❖Welding

- Welding = joining metals by heat (and/ or pressure), usually with a filler metal, forming a fusion between them.
- Fusion welding (melt base metal): Arc welding (electric arc 3000–4000 °C; SMAW/ stick, MIG/ GMAW, TIG/ GTAW, submerged arc) – gas welding (oxy-acetylene flame ≈ 3200 °C) – thermit welding (railways), laser, electron beam.
- Solid-state / pressure welding: resistance welding (spot, seam) – body panels; friction (stir) welding, forge welding, ultrasonic.
- Brazing (filler > 450 °C, base metal does not melt – brass filler for Cu/ steel), Soldering (filler < 450 °C – Pb-Sn, electronics, pipes).
- Arc welding equipment: power source (transformer/ rectifier/ inverter), electrode & holder, earth clamp, cables, helmet (shade), gloves (safety!).
- Weld defects: porosity, slag inclusion, cracks, undercut, incomplete penetration, distortion.
- Applications: ships, pressure vessels, pipelines, bridges, car bodies, structures, machinery repair.
- Advantages: strong, leak-proof joints, light structure, versatile. Disadvantages: residual stress, needs skill, permanent, HAZ (heat-affected zone).

5. Machining

> **Machining** = removal of excess material in the form of chips from a workpiece using a cutting tool on a machine tool to obtain accurate shape, size and surface finish.

❖Cutting parameters

- Cutting speed (V) = π D N / 1000 m/min; Feed (f) – tool advance per revolution/ stroke (mm/rev); Depth of cut (d) (mm). Material removal rate MRR = V f d.

❖Basic machine tools & operations

Machine	Motion (work / tool)	Operations
Lathe	Work rotates, tool moves linear	Turning, facing, drilling, boring, threading, knurling, parting
Drilling machine	Drill rotates & feeds	Drilling, reaming, tapping, counter-sinking
Shaper / Planer	Tool reciprocates (shaper)	Flat surfaces, slots, keyways
Milling machine	Multi-tooth cutter rotates; work feeds	Slots, gears, flat/ contoured surfaces
Grinding machine	Abrasive wheel	Fine finish, tool sharpening
Broaching, sawing, boring		Keyways, cut-off

- Cutting tool materials: HSS, carbides (tungsten carbide), ceramics, CBN, diamond; cutting fluids (coolant & lubricant).
- Tool geometry: rake angle, clearance (relief) angle, nose radius.
- Non-conventional machining: EDM (spark erosion), ECM, USM, laser, abrasive water jet – for hard materials/ complex shapes.
- Finishing: polishing, honing, lapping, buffing.

6. CNC Machines

> **CNC (Computer Numerical Control)** = machine tools whose movements are controlled automatically by a computer executing a pre-programmed set of coded instructions (G-code & M-code).

- **Parts:** Part program (G-code), MCU (machine control unit/ computer), drives & servomotors, feedback (encoders, resolvers), machine tool structure (slides, spindle, ball screws), tool changer (ATC), coolant.
- **Axes:** X, Y, Z (+ A, B, C rotary); 3-axis, 5-axis machining centres.
- **Types:** CNC lathe (turning centre), CNC milling/ VMC & HMC, CNC drilling, grinding, EDM, laser/ plasma cutting, routers, CNC press brake.
- **Open loop vs closed loop control:** closed loop uses feedback → accuracy.
- **G-codes:** G00 rapid move; G01 linear feed; G02/ G03 circular arc CW/ CCW; G90 absolute; G91 incremental. **M-codes:** M03 spindle ON clockwise; M05 stop; M06 tool change; M30 program end.
- **Advantages:** high accuracy & repeatability, complex shapes, less manual skill, reduces setup time, consistent quality, 24 × 7 production, safety.
- **Disadvantages:** high initial & maintenance cost, skilled programmer needed, power dependent.
- **Applications:** automotive & aerospace parts, moulds & dies, medical implants, electronics.

7. 3D Printing (Additive Manufacturing)

> Additive manufacturing (3D printing) builds a part layer by layer from a 3D CAD model by adding material, instead of removing it (subtractive).

❖Process steps

- CAD model → STL file → slicing (layers) → printing → post-processing (support removal, curing, polishing).

❖Technologies

- FDM/ FFF (fused deposition – melted plastic filament: PLA, ABS); SLA (stereolithography – UV laser cures liquid resin); SLS (laser sinters powder – nylon); SLM/ DMLS (metal powder bed – Ti, steel, Inconel); Binder jetting; LOM; Electron beam melting; WAAM; concrete 3D printing (houses).

❖Advantages

- Complex shapes, rapid prototyping, customisation, low waste, no tooling, light-weight lattice structures, on-demand spares.

❖Limitations

- Slow for mass production, limited size/ materials, surface finish & strength anisotropy, high cost for metals.

❖Applications

- Prototyping, aerospace (fuel nozzles), medical (implants, dental, prosthetics), automotive, jewellery, tooling, construction, education.

8. Smart Manufacturing (Industry 4.0)

> Smart manufacturing = using connected digital technologies so that factories sense, analyse, decide and adapt automatically.

- Key technologies: IoT sensors, Big data & AI/ ML, cloud/ edge computing, cyber-physical systems, digital twin (virtual copy of machine/ plant), robots & cobots, AR/ VR, 5G, additive manufacturing, MES/ ERP integration, predictive maintenance.
- Benefits: higher productivity, quality & flexibility, less downtime & waste, energy saving, mass customisation, safety, real-time monitoring.
- Challenges: investment, cyber-security, skilled workforce, interoperability.
- Industrial revolutions: 1.0 steam/ mechanisation → 2.0 electricity/ mass production → 3.0 electronics/ automation (PLC, CNC) → 4.0 cyber-physical/ smart.

9. THERMAL ENGINEERING – Basics

- Thermodynamics: science of energy – heat, work. System, surroundings, closed/ open systems.
- First law: $Q = \Delta U + W$ (energy is conserved). Second law: heat cannot flow from cold to hot by itself; no engine is 100% efficient (Kelvin–Planck); Clausius statement for refrigerators.
- Heat engine: receives heat Q_1 from source, converts part into work W, rejects Q_2 to sink: $\eta = W/Q_1 = 1 - Q_2/Q_1$; Carnot $\eta = 1 - T_2/T_1$.
- Refrigerator: takes heat from cold space using work; $COP = Q_L/W$.

10. Boilers

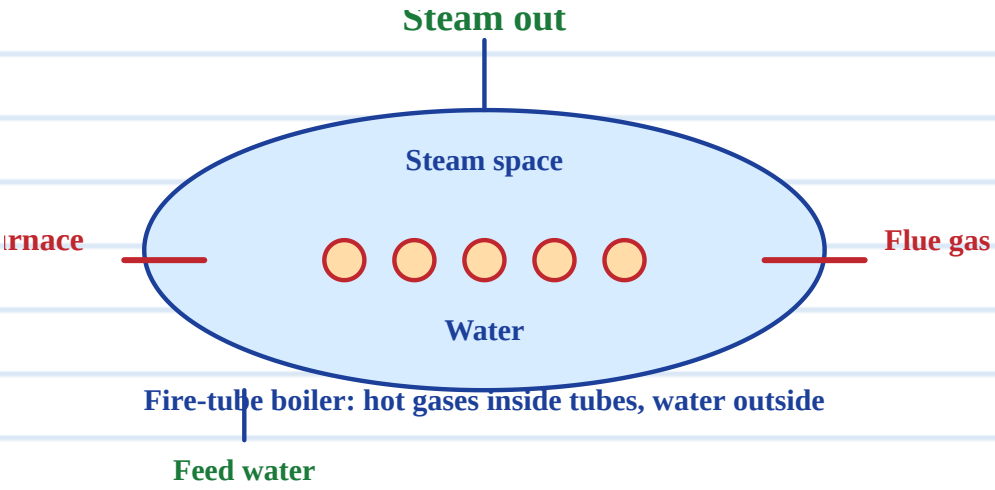


Fig 2.3 Fire-tube boiler (Cochran/ Lancashire principle)

> A boiler (steam generator) is a closed vessel in which water is heated to produce steam at the required pressure and temperature, using heat from fuel combustion.

❖Classification

- Fire-tube boilers: hot gases flow through tubes surrounded by water – simple, low pressure (≤ 20 bar), low capacity; e.g., Cochran, Lancashire, locomotive, Scotch marine.
- Water-tube boilers: water flows in tubes, hot gases outside – high pressure (up to 250 bar), large capacity, quick steam; e.g., Babcock & Wilcox, Stirling; used in power plants.
- Others: vertical/ horizontal, internally/ externally fired, natural/ forced circulation (Benson, La Mont), supercritical.

❖Working principle (fire-tube)

- Fuel burns in furnace → hot flue gases pass through the tubes → heat transferred to surrounding water → water boils → steam collects in steam space → leaves through steam stop valve to the engine/ turbine; feed water pumped in; flue gas exits via chimney.

❖Boiler mountings (safety & control – on the boiler)

- Water level indicator, pressure gauge, safety valve, steam stop valve, feed check valve, blow-off cock, fusible plug, man-hole.

❖Boiler accessories (improve efficiency)

- Economiser (heats feed water using flue gas), air pre-heater, superheater (raises steam temperature above saturation), feed pump, injector.

❖Fire-tube vs water-tube

Feature	Fire-tube	Water-tube
Hot gas	Inside tubes	Outside tubes
Pressure	Low (≤ 20 bar)	High (≤ 250 bar)
Capacity	Small	Large
Steam raising	Slow	Quick
Risk	Heavy explosion	Less (small tubes)
Cost	Low	High

- Boiler efficiency η = (steam generated × $(h_s - h_f)$) / (fuel × CV) ≈ 75–90%. Equivalent evaporation.

II. Internal Combustion (IC) Engines

> IC engine = heat engine in which fuel is burned inside the engine cylinder, and the high-pressure gases push a piston, which turns the crankshaft through a connecting rod.

❖Main components

- Cylinder block, cylinder head, piston, piston rings, gudgeon pin, connecting rod, crankshaft, flywheel, camshaft, inlet & exhaust valves, spark plug (SI) / fuel injector (CI), carburettor/ fuel pump, manifold, crankcase, gaskets, cooling & lubrication systems.

❖Terms

- TDC & BDC (top & bottom dead centre), stroke (L), bore (D), swept (displacement) volume $V_s = (\pi/4)\, D^2 L$, clearance volume V_c, compression ratio $r = (V_s + V_c)/V_c$ (SI 6–10; CI 15–22).

❖Classification

- By fuel: petrol, diesel, gas (CNG/ LPG), bio-fuel; by cycle: Otto, Diesel; by strokes: 2-stroke, 4-stroke; by ignition: SI/ CI; by cylinders: single/ multi; cooling: air/ water; arrangement: inline, V, radial.

12. Four-Stroke Engine

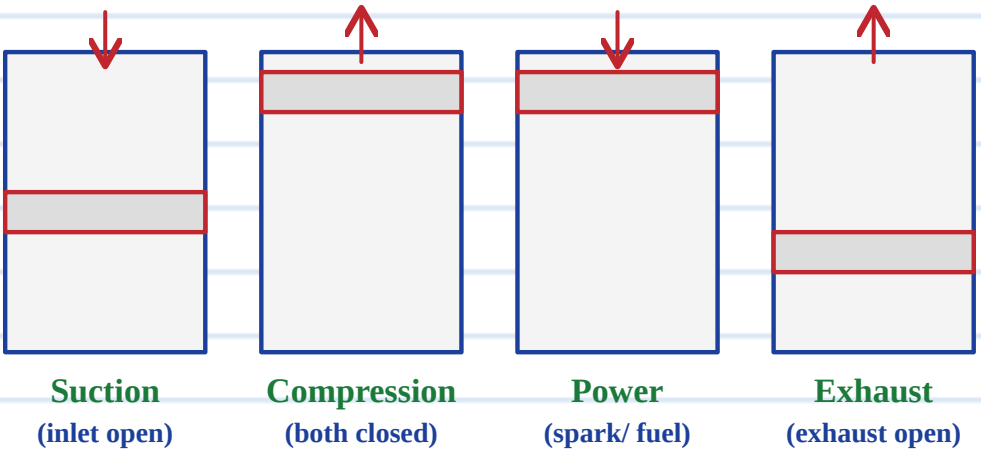


Fig 2.1 Four strokes: piston down-up-down-up (2 revolutions per cycle)

- **Cycle completed in 4 piston strokes (2 crank revolutions = 720°);** one power stroke per 2 revolutions.
- **1. Suction (intake):** piston TDC → BDC; **inlet valve open;** air-fuel mixture (petrol) or only air (diesel) drawn in.
- **2. Compression:** BDC → TDC; both valves closed; charge compressed (petrol: spark at end; diesel: air heated to 500–600 °C).
- **3. Power (expansion):** burning gases push piston TDC → BDC; **only power-producing stroke;** both valves closed.
- **4. Exhaust:** BDC → TDC; **exhaust valve open;** burnt gases expelled.
- Valves operated by **camshaft** (rotates at half crank speed).
- Applications: cars, buses, trucks, tractors, generators.

13. Two-Stroke Engine

- **Cycle completes in 2 strokes (1 revolution = 360°);** one power stroke every revolution. **No valves – ports (inlet, transfer, exhaust)** uncovered by piston.
- **Upward stroke:** piston compresses charge above, simultaneously **fresh charge is drawn into crankcase** (through inlet port). Ignition near TDC.
- **Downward stroke (power):** expansion; piston uncovers exhaust port → burnt gas out; then **transfer port** opens, **crankcase charge pushed into cylinder** (scavenging).
- Applications: mopeds, scooters, chain-saws, outboard motors, small generators.

❖2-stroke vs 4-stroke

Feature	2-stroke	4-stroke
Power strokes	Every revolution	Every 2 revolutions
Valves	Ports	Valves + cam
Power for same size	More (1.5–1.8×)	Less
Fuel economy / thermal efficiency	Lower (charge loss)	Higher
Lubrication	Mixed oil in fuel	Separate sump
Pollution	More	Less
Weight, cost	Light, cheap	Heavier, costly
Smoothness	Less	More

14. SI & CI Engines

Feature	SI (Petrol – Otto)	CI (Diesel)
Fuel	Petrol (high volatility, high octane)	Diesel (high cetane no.)
Mixture formation	Carburettor/ MPFI (outside)	Injector (inside cylinder)
Ignition	Spark plug	Compression (self) ignition
Compression ratio	6 – 10	15 – 22
Cycle	Otto (constant volume heat addition)	Diesel (constant pressure)
Thermal efficiency	25 – 30%	35 – 40%
Weight/ cost	Lighter, cheaper	Heavier, costlier
Noise, vibration	Less	More (knock)
Pollutants	CO, HC	NO_x, PM (soot)
Use	Cars, bikes	Trucks, buses, tractors, ships

- Knocking – SI: auto-ignition of end gas (use high octane); CI: delay period (use high cetane).
- Octane rating (SI), Cetane number (CI); Compression ignition requires T > 250–300 °C self-ignition of diesel.

15. Air-Standard Cycles

❖Otto cycle (constant volume) – petrol engine

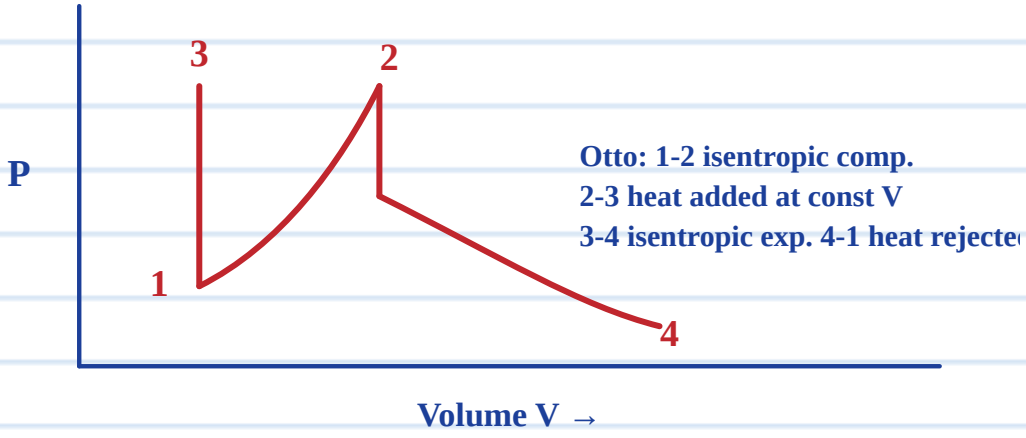


Fig 2.2 P-V sketch of an Otto cycle

- Processes: 1–2 isentropic compression; 2–3 constant volume heat addition (spark); 3–4 isentropic expansion (power); 4–1 constant volume heat rejection.

$$\eta_{Otto} = 1 - 1 / r^{\gamma-1}$$ (r = compression ratio, $\gamma = 1.4$ for air)

- Efficiency increases with r (limited by knocking). Example: $r = 8$: $\eta = 1 - 1/8^{0.4} = 1 - 0.435 = 56.5\%$ (ideal; actual 25–30%).

❖Diesel cycle (constant pressure) – diesel engine

- 1–2 isentropic compression; 2–3 constant pressure heat addition (fuel injection & burning); 3–4 isentropic expansion; 4–1 constant volume heat rejection.

$$\eta_{Diesel} = 1 - [\,1 / r^{\gamma-1}\,] \times [\,(\rho^{\gamma} - 1) / (\gamma\,(\rho - 1))\,]$$, ρ = cut-off ratio = V_3/V_2

- For the same compression ratio, $\eta_{Otto} > \eta_{Diesel}$; but Diesel works at higher r (15–22), so actual diesel engines are more efficient.
- Example: $r = 18$, $\rho = 2$: $\eta = 1 - (1/18^{0.4})[(2^{1.4}-1)/(1.4\times1)] = 1 - 0.315\times[1.639/1.4] = 1 - 0.369 = 63.1\%$.
- Dual cycle – part heat at constant V, part at constant P (modern high-speed diesel).
- Performance: indicated power $IP = p_m L A N n / 60$; brake power $BP = 2\pi NT/60$; mechanical efficiency = BP/IP; brake thermal efficiency = $BP/(\dot{m}_f \times CV)$.

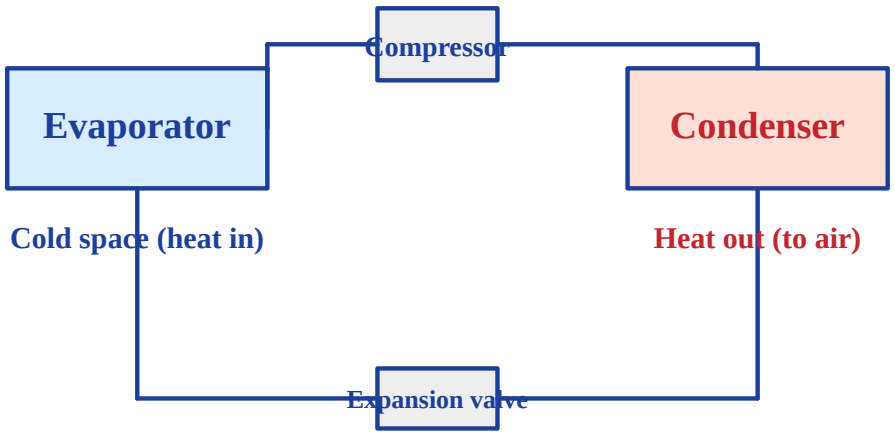
16. Refrigeration

Refrigeration = process of removing heat from a low-temperature space/ substance and rejecting it to a higher-temperature surrounding (needs external work). Device = **refrigerator** (cooling) / **heat pump** (heating).

- **Unit:** 1 ton of refrigeration (TR) = 3.5 kW = 211 kJ/min (heat to freeze 1 tonne of water at 0 °C into ice in 24 h).

COP = Refrigerating effect / Work input = Q_L / W ; Carnot COP = T_L / (T_H – T_L)

❖Vapour Compression Refrigeration (VCR) cycle



Vapour compression refrigeration cycle

Fig 2.4 Vapour compression refrigeration system

- **Compressor:** low-pressure vapour refrigerant → high-pressure, high-temperature vapour (work input).
- **Condenser:** vapour rejects heat to surrounding air/ water and **condenses to liquid.**
- **Expansion valve / capillary tube:** throttling – pressure & temperature drop (partial flash).
- **Evaporator:** cold liquid absorbs heat from the space and **evaporates** → cooling effect; vapour returns to compressor.
- **Refrigerants:** R-134a, R-600a (isobutane), R-290, ammonia (NH_3), CO_2; old CFC/ R-12 banned (**ozone depletion**, Montreal protocol); low GWP preferred.
- **Applications:** domestic refrigerator, cold storage, ice plants, food & medicine preservation, transport.

❖Vapour absorption cycle (brief)

- Uses heat (not work) with NH_3–water or LiBr–water: absorber, pump, generator, condenser, expansion valve, evaporator; suited where waste heat/ solar heat is available; lower COP (~0.6).

17. Air-Conditioning

> Air-conditioning (AC) = process of simultaneously controlling temperature, humidity, cleanliness and distribution of air in a space to meet comfort or process requirements.

- Comfort conditions: temp 22–26 °C, relative humidity 40–60%, air velocity ≈ 0.15–0.25 m/s.

❖Basic processes (psychrometry)

- Sensible cooling/ heating (change in dry bulb temperature); humidification/ dehumidification; cooling & dehumidification; heating & humidification; mixing.
- Psychrometric terms: dry-bulb temp (DBT), wet-bulb temp (WBT), dew point temp (DPT), relative humidity (RH), specific humidity (ω), enthalpy.

❖Working of a window/ split AC

- VCR cycle with the evaporator coil (indoor) cooling & dehumidifying room air blown by a fan; condenser (outdoor) rejects heat; filter cleans air; thermostat; inverter AC varies compressor speed.
- Ratings: tonnage (1.0, 1.5 TR), BEE star rating (ISEER), EER/ COP.

❖Types of AC systems

- Window, split, ductable, VRF/ VRV, chiller (central), packaged unit, evaporative (desert) cooler. Winter AC = heating + humidifying.
- Applications: homes, offices, malls, theatres, hospitals (OT), data centres, trains, vehicles, clean rooms, textile industries.

18. Components of Electric & Hybrid Vehicles

❖Electric Vehicle (EV / BEV)

> An EV uses an electric motor powered by a battery pack (no engine, zero tail-pipe emission).

- Battery pack (Li-ion, LFP/ NMC; 20–100 kWh) – stores energy; BMS (battery management system) monitors voltage, temperature, SOC, balancing.
- Electric motor (BLDC, PMSM, induction) – efficiency ~ 90%+; instant torque.
- Power electronics: inverter (DC → AC for motor), DC–DC converter (for 12 V auxiliaries), motor controller.
- On-board charger (OBC) + charging port (AC slow, DC fast – CCS/ CHAdeMO/ Bharat DC-001).
- Transmission: single-speed reduction gear/ differential (no multi-gear gearbox, no clutch).
- Regenerative braking: motor acts as generator in braking → recovers energy back to battery.
- Thermal management system (battery cooling), vehicle control unit (VCU), display/ telematics, HV cables & contactors, fuse.
- Advantages: zero emissions locally, high efficiency, quiet, low running cost, less maintenance. Limitations: range & charging time, battery cost/ weight, charging infrastructure, recycling.

❖Hybrid Electric Vehicle (HEV)

- Combines IC engine + electric motor + battery.
- Types: Series hybrid (engine only charges battery via generator; motor drives wheels – e.g., range extender); Parallel hybrid (both engine and motor drive wheels); Series–parallel / power-split (Toyota Prius); Mild hybrid (assist only), Full hybrid, Plug-in hybrid (PHEV – external charging).
- Extra components: engine, motor-generator, power-split device/ planetary gear, battery (small), controller, regenerative braking.
- Benefits: 25–40% better mileage, lower emissions, longer range than EV. Limits: complexity & cost.

❖EV vs IC vehicle

Feature	EV	IC engine vehicle
Energy source	Battery	Petrol/ diesel
Emission	Zero tail-pipe	CO_2, NO_x, PM
Efficiency	≈ 85–90%	25–40%
Moving parts	Few	Many
Refuel time	30 min – 8 h	3 min
Noise	Very low	High

- Other: Fuel-cell EV (FCEV) – $H_2 + O_2 \rightarrow$ electricity; India's FAME-II scheme, PLI, National EV mission.

19. Solved Problems

❖Problem 1 – Compression ratio

$D = 100$ mm, $L = 120$ mm, $V_c = 0.1$ L. $V_s = (\pi/4)(0.1)^2(0.12) = 9.42\times10^{-4}$ m^3 = 0.942 L. $r = (0.942 + 0.1)/0.1 = 10.4$.

❖Problem 2 – Brake power

Torque 150 N·m at 3000 rpm: $BP = 2\pi \times 3000 \times 150 / 60{,}000 = 47.1$ kW.

❖Problem 3 – Efficiencies

Fuel 6 kg/h, CV = 42,000 kJ/kg → heat input = 6×42000/3600 = 70 kW; BP = 21 kW → $\eta_{bth} = 30$ %. If IP = 25 kW → $\eta_{mech} = 21/25 = 84$ %.

❖Problem 4 – Refrigerator

$Q_L = 3.5$ kW, $W = 1$ kW → COP = 3.5. Cooling capacity = 1 TR.

❖Problem 5 – Carnot COP

$T_L = 263$ K, $T_H = 303$ K → COP = 263/40 = 6.6 (practical ≈ 3).

20. Important Questions

- Principles of casting, forming, joining (welding) & machining. Write about CNC, 3D printing, smart manufacturing.
- Working of fire-tube/ water-tube boilers; boiler mountings & accessories.
- Draw P–V diagrams and derive efficiency of Otto and Diesel cycles.
- Explain 4-stroke and 2-stroke engines; compare SI and CI engines.
- VCR cycle with a neat sketch; working of air-conditioning.
- Components of EV and hybrid vehicles.

Learn comparison tables – 2-stroke vs 4-stroke, SI vs CI, fire vs water tube – very frequently asked

BASIC MECHANICAL ENGINEERING

Subject Code: 23A01201T (Part B) | UNIT – III : POWER PLANTS, POWER TRANSMISSION & ROBOTICS

1. Power Plants – Introduction

A power plant is an industrial facility that converts a primary energy source (fuel, water, nuclear, wind, sun) into electrical energy, usually through a turbine + generator (alternator) arrangement.

- Sources: coal, oil, gas (thermal); water (hydro); uranium (nuclear); wind, solar, biomass, geothermal (renewable).
- India: installed capacity ≈ 450 GW+; thermal ≈ 55%, renewables ≈ 40%, hydro ≈ 10%, nuclear ≈ 2% (≈ 8 GW).
- Load terms: base load, peak load, load factor = average load / peak load, plant capacity factor, plant efficiency.

Energy conversion chain: Fuel energy → Heat → Mechanical (turbine) → Electrical (generator)

2. Steam (Thermal) Power Plant

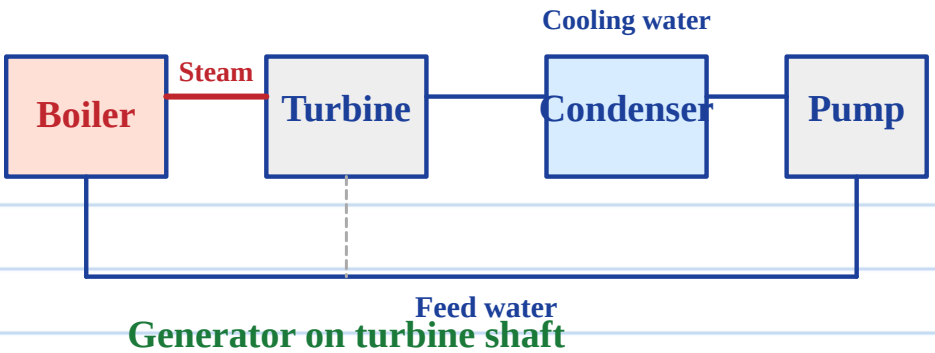


Fig 3.1 Steam (Rankine) power plant – block diagram

Works on the Rankine cycle: fuel (coal) burned in boiler → steam → expands in turbine → generator produces electricity → steam condensed → water pumped back.

❖Main circuits / components

- **Coal & ash handling circuit:** coal storage, crusher, pulveriser (coal → powder), conveyors; ash removed by electrostatic precipitator (ESP), ash pond.
- **Air & flue-gas circuit:** FD fan (forced draught), air pre-heater, furnace, boiler, superheater, economiser, ID fan, chimney (tall stack – disperses gases).
- **Water–steam circuit:** feed pump → economiser → boiler drum (water tubes) → superheater → HP turbine → reheater → IP/ LP turbine → condenser → condensate pump → feed heaters → boiler.
- **Cooling-water circuit:** condenser (cooling water from river/ cooling tower), circulating pump.
- **Turbine + alternator:** steam expands, rotates rotor (3000 rpm for 50 Hz) → alternator → transformer → grid.

❖Working

- 1. Coal pulverised & burnt in boiler furnace (1200–1500 °C). 2. Heat converts water to high pressure steam (160–250 bar, 540–565 °C). 3. Steam passes through turbine → shaft work. 4. Exhaust steam condensed in condenser (vacuum 0.05–0.1 bar) → condensate returns. 5. Generator → electricity at 11–25 kV → stepped up (220/ 400 kV).
- **Efficiency:** overall ≈ 30–40% (supercritical ≈ 42–45%); condenser rejects ≈ 50% heat.
- **Rankine cycle efficiency:** $\eta = (W_T - W_P) / Q_{in}$; improved by superheat, reheat, regeneration (feed heating), high boiler pressure, low condenser pressure.
- **Advantages:** large capacity, reliable, fuel (coal) abundant in India, low cost per kWh, any location near coal/ water. **Disadvantages:** air pollution (CO_2, SO_2, fly ash), huge water need, low efficiency, high initial & running cost, long start-up.
- **Examples:** NTPC Ramagundam, Vijayawada (Narla Tata Rao) TPS, Singareni, Simhadri (A.P.), Talcher, Sipat.

3. Diesel Power Plant

> Uses a **diesel engine (CI engine) coupled to an alternator** to generate electricity – compact, quick-start power for **stand-by, remote, peak load, small towns, hospitals, ships**; capacity 100 kW – 50 MW.

❖Components

- Diesel engine (4-stroke, multi-cylinder, turbocharged, 500–1000 rpm), air intake & filter, turbocharger/ supercharger, fuel storage tank & fuel pump/ injection system (filters), cooling system (radiator/ cooling tower), lubrication system, exhaust system with silencer, starting system (battery/ compressed air), alternator, governor, control panel & switchgear, foundation & vibration isolators.

❖Working

- Air drawn → compressed (~ 30–40 bar, 550 °C) → fuel injected → self-ignites → expansion → power stroke → crankshaft turns alternator → electricity; exhaust to silencer/ waste heat recovery boiler.
- Efficiency 35–45% (best among heat engines).
- Advantages: quick start/ stop (in minutes), small space & foundation, simple, no standby losses, good part-load efficiency, less water, easy installation, suitable for remote places.
- Disadvantages: high fuel & lubricating-oil cost, limited capacity, noise & vibration, higher maintenance, pollution (NO_x, PM), engine life 15–20 yrs.
- Applications: emergency/ stand-by (DG sets in buildings), peak-load, island grids, mobile plants, mines, ships, telecom towers.

4. Hydroelectric Power Plant

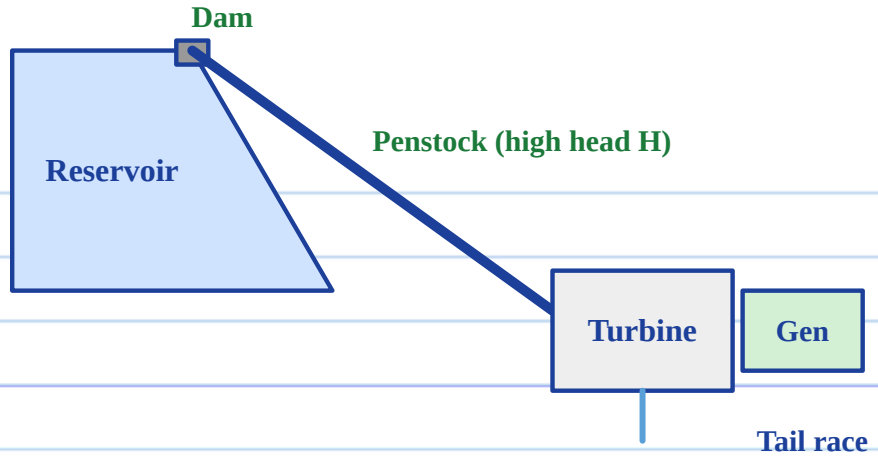


Fig 3.2 Hydroelectric power plant

Converts potential energy of stored/ flowing water (head H) into mechanical energy in a hydraulic turbine, then electrical energy.

$$\text{Power } P = \rho\, g\, Q\, H\, \eta = 9.81 \times Q \times H \times \eta \text{ kW } (Q \text{ in } m^3/s,\ H \text{ in } m)$$

❖Components

- Dam & reservoir (store water, create head), intake/ trash rack/ gates, penstock (pipe carrying water at high pressure), surge tank (absorbs water-hammer), turbine (Pelton – high head > 300 m, impulse; Francis – medium 30–300 m, reaction; Kaplan/ propeller – low head < 30 m, high discharge), draft tube, generator/ alternator, power house, tail race, spillway (flood release), transformers & switch-yard.

❖Working

- Water from reservoir → intake → penstock → hits turbine blades (through nozzle/ guide vanes) → turbine rotates → shaft drives generator → water leaves via draft tube/ tail race to river.
- Classification: by head (low/ medium/ high), by load (base load, peak load, run-of-river), by storage (storage, run-of-river, pumped storage).
- Pumped storage plant: water pumped to upper reservoir during off-peak hours (surplus power) and released during peak demand (Nagarjuna Sagar, Srisailam).
- Efficiency: 85–95%.
- Advantages: renewable, no fuel, no pollution, long life (50–100 yrs), quick start/ stop, low O&M, multipurpose (irrigation, flood control), high efficiency, good for peak load.
- Disadvantages: high initial cost & long construction, depends on rainfall/ site, submergence & rehabilitation, silting, ecological impact, far from load centre (transmission loss).
- Examples: Srisailam (1670 MW), Nagarjuna Sagar, Tehri, Bhakra, Koyna, Idukki.

❖Example

> $Q = 20\ m^3/s$, $H = 100\ m$, $\eta = 0.88$: $P = 9.81 \times 20 \times 100 \times 0.88 = 17{,}266\ kW \approx 17.3\ MW$.

5. Nuclear Power Plant

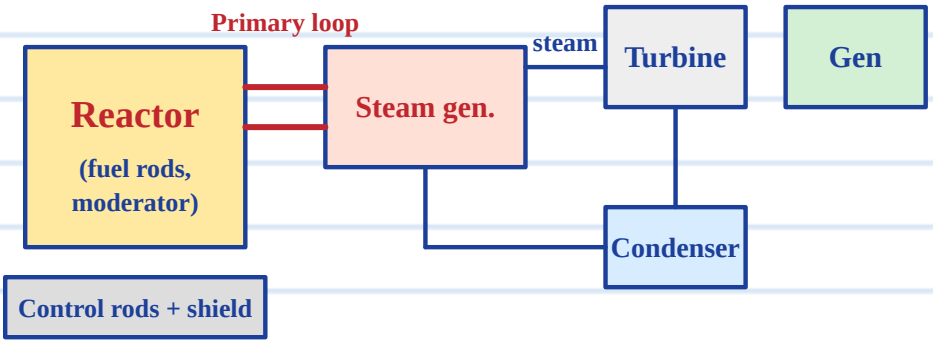


Fig 3.3 Nuclear power plant (PWR layout)

> Heat produced by **nuclear fission** of **U-235** (enriched uranium) is used to make steam which drives a turbine-generator; rest of the plant is similar to a steam plant, with the **reactor replacing the boiler.**

❖Nuclear fission

- A slow (thermal) **neutron** strikes U-235 nucleus → splits into two fission products + **2–3 neutrons** + ~ **200 MeV energy** → neutrons cause further fission → **chain reaction.**

$^{235}U + n \rightarrow$ fission products (Ba, Kr...) + 2.5 n + 200 MeV ; 1 kg U-235 ≈ 2.5×10^6 kg of coal

- **Critical mass**; controlled chain reaction in reactor.

❖Main components of a reactor

- **Fuel:** UO_2 pellets in zircaloy tubes (**fuel rods/ assemblies**); natural U (0.7% U-235) in PHWR; enriched 3–5% in PWR/ BWR.
- **Moderator:** slows fast neutrons to thermal – **heavy water** (D_2O), light water, graphite.
- **Control rods:** Boron/ cadmium/ hafnium – absorb neutrons; inserted/ withdrawn to control power; **SCRAM** = rapid shutdown.
- **Coolant:** carries heat from core – light/ heavy water, CO_2, liquid sodium.
- **Reflector, pressure vessel, biological shield (thick concrete), containment structure.**
- **Steam generator** (PWR/ PHWR), turbine, generator, condenser, cooling tower, waste storage.

❖Types of reactors

- PWR (pressurised water reactor – coolant water at 155 bar does not boil; steam in secondary loop), BWR (boiling water – steam made in core), PHWR (CANDU – heavy water, natural U – India's main type, 220–700 MW), Fast breeder (FBR – Kalpakkam, uses Pu & produce more fuel), gas-cooled, VVER (Kudankulam, with Russia).

❖Working (PWR)

- Fission heats the coolant in primary loop (320 °C, 155 bar) → steam generator transfers heat to secondary water → steam → turbine → generator; condensed & recycled.
- Advantages: huge energy from small fuel, no CO_2, high base load, low fuel cost, reliable. Disadvantages: radioactive waste & disposal, accident risk (Chernobyl, Fukushima), very high capital cost, long construction, decommissioning, safeguards & security.
- India nuclear stations: Tarapur, Rawatbhata (Kota), Kalpakkam, Narora, Kakrapar, Kaiga, Kudankulam; 3-stage programme (U → Pu → Th).

6. Comparison of Power Plants

Feature	Steam	Diesel	Hydro	Nuclear
Fuel	Coal	Diesel oil	Water	Uranium
Efficiency	30–40%	35–45%	85–95%	30–35%
Initial cost	Medium	Low	Very high	Very high
Running cost	Medium–high	High	Very low	Low
Pollution	High	Medium	Nil	Radioactive waste
Start-up	Slow (hours)	Very fast	Fast	Very slow
Capacity	Large	Small	Medium–large	Large
Use	Base load	Standby/ peak	Peak/ base	Base load

7. Mechanical Power Transmission

Power transmission = transferring mechanical power from a driver (motor/ engine shaft) to a driven machine with the required speed and torque, through belts, ropes, chains, gears, shafts, couplings.

Power $P = T\omega = 2\pi N T / 60$ (W) ; Speed ratio = N_1/N_2

- Selection depends on centre distance, speed ratio, power, slip allowed, cost, space.

Drive	Centre distance	Velocity ratio	Slip
Belt / Rope	Large (up to 10 m)	Not exact	Yes
Chain	Medium	Exact (constant)	No
Gear	Short	Exact	No

8. Belt Drives

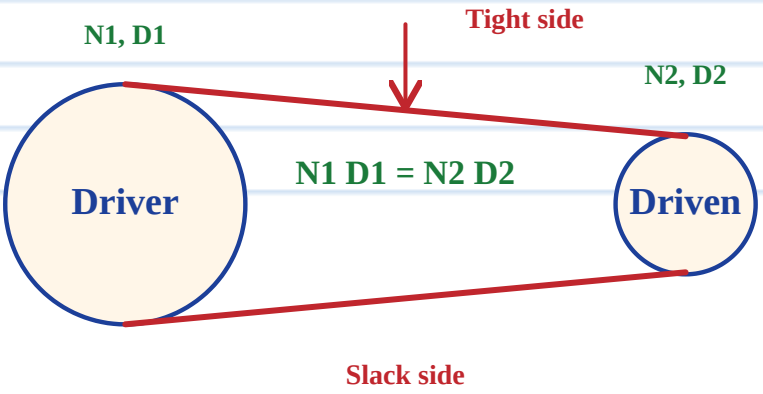


Fig 3.4 Open belt drive

> Belt drive transmits power between two parallel shafts by friction between a flexible belt and pulleys.

❖Types of belts

- Flat belt (leather, rubber, fabric) – large distance, light; V-belt (trapezoidal – higher grip, groove pulley, multiple belts; automobiles, compressors, fans); circular/ round belt (light work, sewing machine); timing (toothed) belt – positive, no slip (car engine cam drive, 3D printers).

❖Arrangements

- Open belt drive: both pulleys rotate in same direction; cross belt drive: opposite direction (more wrap, more wear).
- Others: compound, quarter-twist (non-parallel shafts), stepped cone pulley (speed change), fast & loose pulley.

❖Velocity ratio

$V = \pi D_1 N_1 / 60$ (belt speed) ; $N_1 D_1 = N_2 D_2$ (no slip, thin belt) ; with slip s%: $N_2/N_1 = (D_1/D_2)(1 - s/100)$

- Belt tensions: tight side T_1, slack side T_2; $T_1/T_2 = e^{\mu\theta}$ (flat belt); V-belt: $e^{\mu\theta/\sin\beta}$. Power = $(T_1 - T_2)\,V$.
- Length of open belt: $L \approx \pi(r_1+r_2) + 2C + (r_1-r_2)^2/C$.

❖ Advantages / limitations

- Simple, cheap, smooth & silent, can run over long distances, overload protection (slip), absorbs shock. But slip & creep (velocity ratio not exact), needs tensioning, short life, limited power/ speed, bearing loads high.
- Applications: machine tools, conveyors, fans, pumps, compressors, automobile (alternator, AC, water pump), flour mills, agricultural machines.

9. Chain Drives

Chain drive transmits power by meshing of a chain (links) with toothed wheels (sprockets); positive drive (no slip), used where shafts are parallel and fairly near.

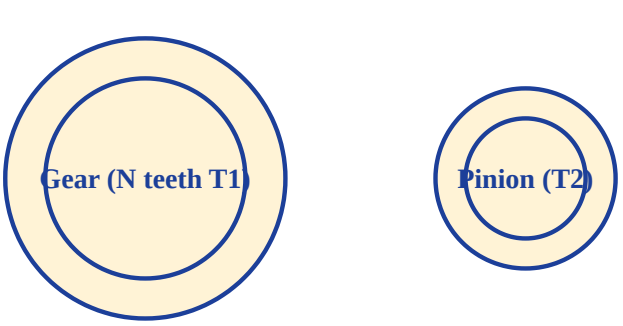
- Types of chains: roller chain (most common – bicycles, motorcycles), silent (inverted-tooth) chain (quiet, high speed), block chain (slow). Pitch = distance between pin centres.
- Velocity ratio: $N_1/N_2 = T_2/T_1$ (T = number of teeth on sprockets).
- Chain speed: $v = N\,T\,p\,/60$. Efficiency ≈ 95–98%.
- Advantages: no slip – exact velocity ratio; high efficiency; transmits large power; compact; works in harsh conditions (dust, heat); no initial tension; long life; drives multiple shafts.
- Disadvantages: costlier, needs lubrication & alignment, noisy at high speed, chordal action/ vibration, elongation (wear), not suitable for non-parallel shafts.
- Applications: bicycles, motorcycles, conveyors, agricultural machinery, timing drive (engine), cranes, rolling mills, textile & printing machines.

10. Rope Drives

Rope drive – power transmitted over long distances (up to 100–150 m) using ropes (fibre ropes – cotton, manila, hemp; or wire ropes) running in grooved pulleys (sheaves).

- Types: fibre rope (multiple ropes in V-groove – mill transmission, 45° groove), wire rope (steel – cranes, lifts, ropeways).
- Systems: multiple (English) system – separate ropes; continuous (American) system – single rope winds many turns with tensioning pulley.
- Tension ratio: $T_1/T_2 = e^{\mu\theta/\sin\beta}$ (β = half groove angle).
- Advantages: long distance, smooth, can transmit large power (many ropes), reliable, less affected by shocks, individual rope failure does not stop drive, quiet.
- Disadvantages: needs large pulleys, ropes wear, creep & slip, higher cost & space for installation, not used for exact timing.
- Applications: cranes, hoists, elevators, ropeways/ cable cars, mining haulage, winches, ski-lifts, textile mills (older).

II. Gear Drives



Speed ratio = N1/N2 = T2/T1

Fig 3.5 Spur gear pair

> Gear = toothed wheel that meshes with another gear to transmit motion and power positively (no slip) between shafts at short centre distance; can change speed, torque and direction.

❖Terms

- Pitch circle (imaginary rolling circle), pitch circle diameter D, module m = D/T (mm), number of teeth T, circular pitch p = πm, addendum, dedendum, pressure angle (20°), backlash, face width, pinion = smaller gear, gear = larger gear.

Velocity (speed) ratio = $N_1/N_2 = D_2/D_1 = T_2/T_1$; Centre distance = $(D_1+D_2)/2 = m(T_1+T_2)/2$

- Gear train – series of gears: simple (each shaft one gear), compound, reverted, planetary (epicyclic); overall ratio = product of stage ratios; torque ↑ when speed ↓ ($T_1N_1 = T_2N_2$ for power).

❖Types of gears

Gear type	Shafts	Features & uses
Spur	Parallel	Straight teeth, simple, noisy; gearboxes, clocks
Helical	Parallel	Slanted teeth, smooth, quiet, axial thrust; car transmissions
Double helical (herringbone)	Parallel	No axial thrust; turbines, heavy rolling mills
Bevel (straight/ spiral)	Intersecting (90°)	Cone gears; differential, hand drills
Worm & worm wheel	Non-intersecting (90°)	High reduction (up to 100:1), self-locking; lifts, conveyors, steering
Rack & pinion	Rotary ↔ linear	Steering, lathe carriage
Internal gear / planetary	Concentric	Compact, high ratio; automatic transmissions

- Gear materials: steel, cast iron, brass/ bronze (worm wheel), nylon/ plastic. Lubrication (oil bath, splash) & cooling essential.
- Advantages: positive drive, exact velocity ratio, high efficiency (95–99%), compact, wide range of power, long life, constant torque transmission.
- Disadvantages: costly manufacture, needs accurate alignment & lubrication, noise at high speed, cannot be used for long distances, shock load sensitive.
- Applications: automobile gearbox & differential, machine tools, clocks/ watches, wind turbine gearbox, robots, lifts, steering, mixers, cement mills.

❖Comparison of drives

Feature	Belt	Chain	Rope	Gear
Drive type	Friction	Positive	Friction	Positive
Slip	Yes	No	Yes	No
Distance	Large	Medium	Very large	Small
Efficiency	92–96%	95–98%	90%	95–99%
Noise	Silent	Medium	Silent	Noisy
Lubrication	No	Yes	No	Yes
Cost	Low	Medium	Medium	High

12. Introduction to Robotics

A robot is a programmable, multi-functional machine/ manipulator that can sense, think (control) and act, performing tasks automatically (ISO: "automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes"). Robotics = design, build, operate & use robots. Word "robot" (Čapek 1920); Asimov's 3 laws.

❖Components of a robot

- Manipulator/ mechanical structure (links & joints), end-effector (gripper/ tool), actuators (electric servo/ stepper motors, hydraulic, pneumatic), sensors (position encoders, force/ torque, vision, proximity, LiDAR), controller (brain – computer/ PLC), power supply, software/ AI, teach pendant/ HMI.

❖Joints & links

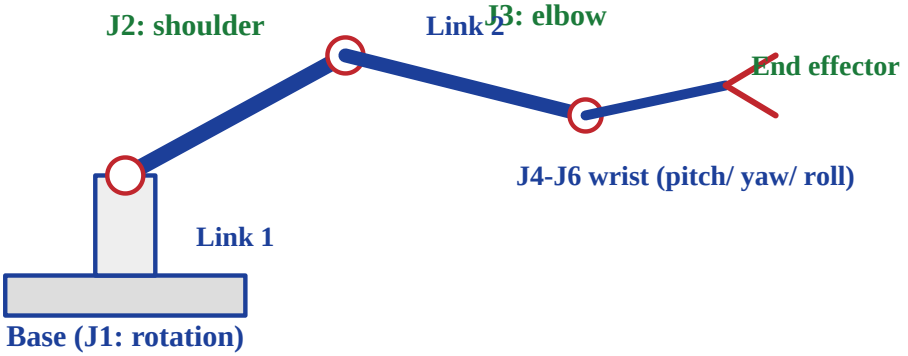


Fig 3.6 Articulated robot arm: joints connect rigid links

- Link = rigid member of the arm; Joint = connection between links providing relative motion (a degree of freedom, DOF). DOF = number of independent motions; 6 DOF needed for arbitrary position (3) + orientation (3).
- Types of joints:
  - Revolute (R) / rotary – rotation about an axis (elbow).
  - Prismatic (P) / linear – sliding translation (telescoping).
  - Cylindrical (R + P), Spherical (3 rotations, ball-socket), Screw/ helical, Planar.
- Notation: e.g. RRR, PRR, PPP.
- Wrist joints: roll, pitch, yaw (orient the end-effector).

- **End-effectors:** mechanical grippers, vacuum/ suction cups, magnetic grippers, welding torch, spray gun, drill, soft grippers.

❖Robot configurations (by arm geometry)

Configuration	Joints	Work envelope	Examples / uses
Cartesian (gantry)	P-P-P	Cuboid	CNC, 3D printers, pick & place
Cylindrical	R-P-P	Cylinder	Assembly, handling
Spherical (polar)	R-R-P	Sphere section	Welding, die casting
SCARA (selective compliance articulated)	R-R-P	Cylinder-like	Fast electronics assembly
Articulated (revolute)	R-R-R	Large irregular	Welding, painting, material handling – most common (6-axis)
Delta/ parallel	Parallel links	Dome	High-speed packaging

- **Other types:** mobile robots (wheeled, legged, AGV), humanoids, drones (UAV), cobots (collaborative), underwater (ROV), surgical, exoskeleton.
- **Key specifications:** payload, reach/ work envelope, repeatability (± 0.02 mm), speed, DOF, accuracy.

❖Applications of robotics

- **Manufacturing:** welding, painting/ spraying, assembly, pick-and-place, machine tending, inspection, packaging, palletising (automotive plants: ~ 90% of car body welding by robots).
- **Medical:** surgery (da Vinci), rehabilitation, prosthetics, pharmacy automation.
- **Space & defence:** Mars rovers (Perseverance, Chandrayaan Pragyan), bomb disposal, surveillance drones.
- **Agriculture:** harvesting, spraying, milking. **Logistics:** warehouse robots (Amazon), AGVs, delivery drones. **Hazardous tasks:** nuclear, mining, firefighting. **Home/ service:** robot vacuum, hotel & restaurant robots, education, entertainment.
- **Advantages:** accuracy, repeatability, speed, 24×7 work, safety in dangerous tasks, quality, reduce cost per unit. **Limitations:** high investment, programming skill, job displacement, limited flexibility/ intelligence, maintenance.

13. Solved Problems

❖Problem 1 – Belt drive

Driver D_1 = 300 mm at 600 rpm; driven D_2 = 600 mm. N_2 = 600×300/600 = **300 rpm**. With 2% slip: N_2 = 300 × 0.98 = **294 rpm**.

❖Problem 2 – Belt power

T_1 = 1200 N, T_2 = 400 N, belt speed V = 10 m/s → P = (1200 – 400) × 10 = **8 kW**.

❖Problem 3 – Gears

Pinion 20 teeth at 1200 rpm drives gear of 60 teeth: N_2 = 1200×20/60 = **400 rpm**; speed ratio 3:1; torque on gear is 3× pinion torque. If module = 4 mm: centre distance = 4×(20+60)/2 = **160 mm**.

❖Problem 4 – Chain drive

Sprockets 15 teeth (driver, 90 rpm) & 45 teeth (driven): N_2 = 90×15/45 = **30 rpm**.

❖Problem 5 – Nuclear energy

1 kg U-235 complete fission ≈ 8.2 × 10^{13} J ≈ **22.8 million kWh**.

❖Problem 6 – Steam plant

Plant output 500 MW at 38% efficiency → heat needed = 500/0.38 = 1316 MW; coal (CV 20 MJ/kg) = 1316/20 = **65.8 kg/s ≈ 237 tonnes/hour**.

❖Problem 7 – DOF

Arm with 6 revolute joints has **6 DOF** (3 for position, 3 for orientation).

14. Important Questions

- Explain with a neat sketch the working of a steam power plant / diesel / hydro / nuclear power plant.
- Compare steam, diesel, hydro and nuclear plants.
- Explain belt drive, chain drive, rope drive and gear drive with applications; types of belts & gears.

- Define robot, joints, links and DOF. Explain robot configurations and applications.

Revise: $P = \rho g Q H \eta$; $N_1 D_1 = N_2 D_2$; $N_1/N_2 = T_2/T_1$